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1 Basis

1.1 Default Code

```
// g++args: -std=c++17 -Wall -Wshadow -fsanitize=undefined
#pragma GCC optimize("O3,unroll-loops")
#pragma target optimize("avx2,bmi,bmi2,lzcnt,popcnt")
```

1.2 Some Useful Primes

```
/* 13331, 75577, 999983, 1097774749, 1076767633,
 * 100102021, 999997771, 1001010013, 1000512343,
 * 987654361, 999991231, 999888733, 98789101, 987777733,
 * 999991921, 1010101333, 1010102101, 1000000000039,
 * 1000000000000037, 2305843009213693951,
 * 4611686018427387847, 9223372036854775783,
 * 18446744073709551557 */
```

1.3 偽隨機 & MT19937

```
// chrono::steady_clock::now().time_since_epoch().count();
mt19937 gen(*seed*/ hash<string>("ShanC_orz"));
mt19937_64 mt(*seed*/ 114514);
int randint(int a, int b) { // [a, b]
    return uniform_int_distribution<int>(a, b)(gen);
}
// srand(time(NULL)), rand() % (b - a + 1) + a;
```

1.4 快讀快寫

```
inline int read() {
    char c = getchar();
    int x = 0, f = 1;
    while (c < '0' || c > '9') {
        if (c == '-') f = -1;
        c = getchar();
    }
    while (c >= '0' && c <= '9')
        x = x * 10 + c - '0', c = getchar();
    return x * f;
}
inline void write(int x) {
    if (x < 0) putchar('-'), x = -x;
    if (x > 9) write(x / 10);
    putchar(x % 10 + '0');
}
```

1.5 怪怪的東西

```
#define SECs ((double)clock() / CLOCKS_PER_SEC)
struct KeyHasher {
    size_t operator()(const Key& k) const {
        return k.first + k.second * 100000;
    }
};
typedef unordered_map<Key, int, KeyHasher> map_t;
__builtin_popcountll; // 二進位有幾個1
__builtin_clzll; // 左起第一個1之前0的個數
__builtin_parityll; // popcount的奇偶性
__builtin_mul_overflow(a, b, &h) // a*b是否溢位
__lg(x); // floor(log2(x))
```

2 Data Structures

2.1 Disjoint Set (併查集)

```
struct DSU {
    int n;
    vector<int> f, sz;
    DSU(int _n) : n(_n) { // 初始化
        f.resize(n);
        sz.resize(n);
        for (int i = 0; i < n; i++) f[i] = i, sz[i] = 1;
    }
    int find(int x) { // 返回 x 的根節點
        if (x == f[x]) return x;
        return f[x] = find(f[x]);
    }
    void merge(int x, int y) { // 將 x 和 y 合併
        x = find(x), y = find(y);
        if (x == y) return;
        if (sz[x] < sz[y]) swap(x, y); // 將小的併入大的
        sz[x] += sz[y];
        f[y] = x;
    }
};
```

2.2 Binary Indexed Tree

```
struct BIT {
    int n;
    vector<ll> bit;
    int lowbit(int x) { return x & -x; }
    BIT(int _n) : n(_n + 1) { bit = vector<ll>(_n + 1, 0); }
    void update(int x, ll v) { // 將 x 的值加上 v
        for (; x < n; x += lowbit(x)) bit[x] += v;
    }
    ll query(int x) { // 查詢區間 [1, x] 的總和
        ll ret = 0;
        for (; x; x -= lowbit(x)) ret += bit[x];
        return ret;
    }
    ll query(int l, int r) { // 查詢區間 [l, r] 的總和
        return query(r) - query(l - 1);
    }
    int kth(int k) {
        int x = 0;
        for (int i = 1 << __lg(n); i; i >= 1)
            if (x + i <= n && k >= bit[x + i - 1])
                x += i, k -= bit[x - 1];
        return x;
    }
};
```

```

}
};

2.3 二維 BIT

struct BIT {
    int n;
    vector<vector<ll>>> bit;
    int lowbit(int x) { return x & -x; }
    BIT(int _n) : n(_n + 1) {
        bit.assign(n, vector<ll>(n, 0));
    }
    void update(int x, int y, ll val) { // 更新 (x, y)
        for (int i = x; i < n; i += lowbit(i))
            for (int j = y; j < n; j += lowbit(j))
                bit[i][j] += val;
    }
    ll query(int x, int y) { // 查詢 (1, 1) ~ (x, y) 的和
        ll ans = 0;
        for (int i = x; i; i -= lowbit(i))
            for (int j = y; j; j -= lowbit(j)) ans += bit[i][j];
        return ans;
    }
    ll range_query(int a, int b, int x, int y) {
        return query(x, y) - query(x, b - 1) -
            query(a - 1, y) + query(a - 1, b - 1);
    }
};

```

2.4 Segment Tree

```

#define cl (i << 1)
#define cr (i << 1 | 1)
#define NO_TAG 0
struct SegmentTree { // 1-base
    int n;
    vector<int> seg, tag;
    SegmentTree(int _n) : n(_n) { // 空的 SegmentTree
        seg.resize(n * 4), tag.resize(n * 4);
    }
    void push(int i, int l, int r) {
        if (tag[i] != NO_TAG) {
            seg[i] += tag[i]; // update by tag
            if (l != r)
                tag[cl] += tag[i], tag[cr] += tag[i]; // push
            tag[i] = 0;
        }
    }
    void pull(int i, int l, int r) {
        int mid = (l + r) >> 1;
        push(cl, l, mid), push(cr, mid + 1, r);
        seg[i] = max(seg[cl], seg[cr]); // pull (操作改這)
    }
    void build(int i, int l, int r) {
        if (l == r) {
            seg[i] = arr[l]; // 初始值
            return;
        }
        int mid = (l + r) >> 1;
        build(cl, l, mid), build(cr, mid + 1, r);
        pull(i, l, r);
    }
    void update(int i, int l, int r, int ql, int qr,
        int v) {
        push(i, l, r);
        if (ql <= l && r <= qr) {
            tag[i] += v;
            return;
        }
        int mid = (l + r) >> 1;
        if (ql <= mid) update(cl, l, mid, ql, qr, v);
        if (qr > mid) update(cr, mid + 1, r, ql, qr, v);
        pull(i, l, r);
    }
    int query(int i, int l, int r, int ql, int qr) {
        push(i, l, r);
        if (ql <= l && r <= qr) return seg[i];
        int mid = (l + r) >> 1, ans = -INF;
        if (ql <= mid) // (操作改這)
            ans = max(ans, query(cl, l, mid, ql, qr));
        if (qr > mid) // (操作改這)
            ans = max(ans, query(cr, mid + 1, r, ql, qr));
        return ans;
    }
};

```

```

}
// 區間 [ql, qr] 加值 v
void update(int ql, int qr, int v) {
    update(1, 1, n, ql, qr, v);
}
// 查詢區間 [ql, qr]
int query(int ql, int qr) {
    return query(1, 1, n, ql, qr);
}
};

```

2.5 Treap

```

struct Treap {
    int sz, val, pri, tag;
    Treap *l, *r;
    Treap(int _val) : sz(1), val(_val), tag(0) {
        l = r = NULL, pri = rand(); // 隨機產生優先度
    }
};
int size(Treap* a) { return a ? a->sz : 0; }
void pull(Treap* a) {
    a->sz = size(a->l) + size(a->r) + 1;
}
Treap* merge(Treap* a, Treap* b) {
    // val of a is always bigger than val of b
    if (!a || !b) return a ? a : b;
    if (a->pri > b->pri) {
        a->r = merge(a->r, b);
        pull(a);
        return a;
    } else {
        b->l = merge(a, b->l);
        pull(b);
        return b;
    }
}
void split(Treap* t, int k, Treap*& a, Treap*& b) {
    // a<k, b>=k
    if (!t) {
        a = b = NULL;
        return;
    }
    if (k <= t->val) {
        b = t;
        split(t->l, k, a, b->l);
        pull(b);
    } else {
        a = t;
        split(t->r, k, a->r, b);
        pull(a);
    }
}
Treap* add(Treap* t, int v) {
    Treap *val = new Treap(v), *l = NULL, *r = NULL;
    split(t, v, l, r);
    return merge(merge(l, val), r);
}
Treap* del(Treap* t, int v) {
    Treap *l, *mid, *r, *temp;
    split(t, v, l, temp);
    split(temp, v + 1, mid, r);
    return merge(l, r);
}
int position(Treap* t, int p) { // base 1
    if (size(t->l) + 1 == p) return t->val;
    if (size(t->l) < p)
        return position(t->r, p - size(t->l) - 1);
    return position(t->l, p);
}
int query(Treap* t, int k) { // num of >= k
    if (!t) return 0;
    if (t->val == k) return size(t->l) + 1;
    if (t->val > k) return query(t->l, k);
    return size(t->l) + 1 + query(t->r, k);
}

```

2.6 平板電腦

```

#include <bits/stdc++.h>

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/rope>
#include <functional>

```

```
#include <queue>
using namespace std;
using namespace __gnu_cxx;
using namespace __gnu_pbds;
typedef tree<int, null_type, less<int>, rb_tree_tag,
            tree_order_statistics_node_update>
    set_t;
typedef cc_hash_table<int, int> umap_t;
typedef priority_queue<int> heap;

set_t s; // Insert some entries into s.
s.insert(12), s.insert(505);
// The order of the keys should be: 12, 505.
assert(*s.find_by_order(0) == 12);
assert(*s.find_by_order(3) == 505);
// The order of the keys should be: 12, 505.
assert(s.order_of_key(12) == 0);
assert(s.order_of_key(505) == 1);
s.erase(12); // Erase an entry.
// The order of the keys should be: 505.
assert(*s.find_by_order(0) == 505);
// The order of the keys should be: 505.
assert(s.order_of_key(505) == 0);
heap h1, h2;
h1.join(h2);
rope<char> r[2];
r[1] = r[0]; // persistenet
string t = "abc";
r[1].insert(0, t.c_str());
r[1].erase(1, 1);
cout << r[1].substr(0, 2);
```

3 Mathematics

3.1 公式們

$$\sum_{k=1}^n n = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n n^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

$$\sum_{k=1}^n n^4 = n(1+n)(1+2n)(-1+3n+3n^2)/30$$

$$\sum_{k=1}^n n^5 = n^2(n+1)^2(2n^2+2n-1)/12$$

$$C_n = \frac{C_n^{2n}}{n+1} = \frac{(2n)!}{n! * (n+1)!}$$

錯位排列

$$DP_i = \begin{cases} 0 \vee 1, & i = 1 \vee 2 \\ (i-1)(DP_{i-1} \times DP_{i-2}), & i \geq 3 \\ iDP_{i-1} + (-1)^i & \text{另一種關係式} \end{cases}$$

生成樹數量

$$n^{n-2}$$

$$P_k^n = \frac{n!}{k!(n-k)!}, \quad C_k^n = \frac{n!}{(n-k)!}, \quad H_k^n = C_{n-1}^{n+k-1}$$

$$\pi = \arccos(-1), \quad e^{i\theta} = \cos \theta + i \sin \theta, \quad pV = nTR$$

$$a^{\varphi(m)} \equiv 1 \pmod m, \quad a^b \equiv a^{b \bmod \varphi(m)-1} \pmod m$$

$$\varphi(m) = m-1, \text{When } m \text{ is Prime}$$

$$\mu(n) = \begin{cases} 1, & n = 1 \\ (-1)^k, & n = p_1 p_2 \cdots p_k \text{ (沒平方項)}, \\ 0, & \text{otherwise} \end{cases}, \quad (f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

$$\mu * 1 = \epsilon \quad \mu * f = g \quad \varphi = \mu * \text{Id} \quad \varphi * 1 = \text{Id}$$

$$\sum_{i=1}^N \sum_{j=1}^M [\gcd(i, j) = 1] = \sum_{d=1}^N \mu(d) \left\lfloor \frac{M}{d} \right\rfloor \left\lfloor \frac{N}{d} \right\rfloor$$

$$\sum_{i=1}^N \sum_{j=1}^M \gcd(i, j) = \sum_{T=1}^{\min(N, M)} \phi(T) \lfloor \frac{N}{T} \rfloor \lfloor \frac{M}{T} \rfloor$$

3.2 階乘 & 模逆元

```
ll fac[MXN], inv[MXN];
void init() {
    fac[0] = 1; // 0! = 1
    for (ll i = 1; i < MXN; i++)
        fac[i] = fac[i - 1] * i % MOD; // n! = n * (n-1)!
    inv[MXN - 1] = power(fac[MXN - 1], MOD - 2);
    for (ll i = MXN - 2; i >= 0; i--)
        inv[i] = inv[i + 1] * (i + 1) % MOD; // (n!)^(-1)
}
```

3.3 GCD & LCM

```
ll gcd(ll a, ll b) {
    if (b && a) // 輾轉先除法 -> 交叉取餘數直到不能再取
        while ((a %= b) && (b %= a)) {}
    return a + b;
}
ll lcm(ll a, ll b) { return a * b / gcd(a, b); }
```

3.4 EXGCD

$ax + by = gcd(a, b)$

```
int exgcd(int a, int b, ll& x, ll& y) {
    if (b == 0) {
        x = 1, y = 0;
        return a;
    }
    int now = exgcd(b, a % b, y, x);
    y -= a / b * x;
    return now;
}
```

3.5 Fast Power

```
ll power(ll x, ll y, ll mod) {
    ll ret = 1;
    while (y) {
        if (y & 1) ret = ret * x % mod;
        x = x * x % mod, y >>= 1;
    }
    return ret;
}
```

3.6 矩陣乘法

```
struct Mat { // Mat t(r, c);
    ll a[200][200], r, c;
    Mat(int _r, int _c) : r(_r), c(_c) {
        memset(a, 0, sizeof(a));
    }
    void build() { // 單位矩陣
        for (int i = 0; i < r; ++i) a[i][i] = 1;
    }
};
Mat operator*(Mat x, Mat y) {
    Mat z(x.r, y.c);
    for (int i = 0; i < x.r; ++i)
        for (int j = 0; j < x.c; ++j)
            for (int k = 0; k < y.c; ++k)
                (z.a[i][j] += x.a[i][k] * y.a[k][j] % MOD) %= MOD;
    return z;
}
```

3.7 FFT

使用前要先跑過 pre_fft(); n 必須是 2^k。

```
const int MAXN = 262144 << 1; // (must be 2^k)
// before any usage, run pre_fft() first
// typedef long double ld;
typedef complex<ld> cplx; // real(), imag()
const ld PI = acos(-1);
const cplx I(0, 1);
cplx omega[MAXN + 1];
void pre_fft() {
    for (int i = 0; i <= MAXN; i++)
        omega[i] = exp(i * 2 * PI / MAXN * I);
}
void fft(int n, cplx a[], bool inv = false) {
    int basic = MAXN / n;
    int theta = basic;
    for (int m = n; m >= 2; m >>= 1) {
        int mh = m >> 1;
        for (int i = 0; i < mh; i++) {
            cplx w = omega[inv ? MAXN - (i * theta % MAXN) : i * theta % MAXN];
            for (int j = i; j < n; j += m) {
                int k = j + mh;
                cplx x = a[j] - a[k];
                a[j] += a[k];
                a[k] = w * x;
            }
        }
        theta = (theta * 2) % MAXN;
    }
}
```

```

int i = 0;
for (int j = 1; j < n - 1; j++) {
    for (int k = n >> 1; k > (i ^= k); k >>= 1) {}
    if (j < i) swap(a[i], a[j]);
}
if (inv)
    for (i = 0; i < n; i++) a[i] /= n;
}
cplx arr[MAXN + 1];
inline void mul(int _n, ll a[], int _m, ll b[],
               ll ans[]) {
    int n = 1, sum = _n + _m - 1;
    while (n < sum) n <= 1;
    for (int i = 0; i < n; i++) {
        double x = (i < _n ? a[i] : 0),
               y = (i < _m ? b[i] : 0);
        arr[i] = complex<double>(x + y, x - y);
    }
    fft(n, arr);
    for (int i = 0; i < n; i++) arr[i] = arr[i] * arr[i];
    fft(n, arr, true);
    for (int i = 0; i < sum; i++)
        ans[i] = (long long int)(arr[i].real() / 4 + 0.5);
}

```

3.8 NTT

```

// Remember coefficient are mod P
/* p=a*2^n+1
   n-----2^n-----p-----a-----root
   16_65536_65537_1_3
   20_1048576_7340033_7_3 */
// (must be 2^k)
template <LL P, LL root, int MAXN>
struct NTT {
    static LL bigmod(LL a, LL b) {
        LL res = 1;
        for (LL bs = a; b; b >>= 1, bs = (bs * bs) % P)
            if (b & 1) res = (res * bs) % P;
        return res;
    }
    static LL inv(LL a, LL b) {
        if (a == 1) return 1;
        return (((LL)(a - inv(b % a, a)) * b + 1) / a) % b;
    }
    LL omega[MAXN + 1];
    NTT() {
        omega[0] = 1;
        LL r = bigmod(root, (P - 1) / MAXN);
        for (int i = 1; i <= MAXN; i++)
            omega[i] = (omega[i - 1] * r) % P;
    }
    // n must be 2^k
    void tran(int n, LL a[], bool inv_ntt = false) {
        int basic = MAXN / n, theta = basic;
        for (int m = n; m >= 2; m >>= 1) {
            int mh = m >> 1;
            for (int i = 0; i < mh; i++) {
                LL w = omega[i * theta % MAXN];
                for (int j = i; j < n; j += m) {
                    int k = j + mh;
                    LL x = a[j] - a[k];
                    if (x < 0) x += P;
                    a[j] += a[k];
                    if (a[j] > P) a[j] -= P;
                    a[k] = (w * x) % P;
                }
            }
            theta = (theta * 2) % MAXN;
        }
        int i = 0;
        for (int j = 1; j < n - 1; j++) {
            for (int k = n >> 1; k > (i ^= k); k >>= 1);
            if (j < i) swap(a[i], a[j]);
        }
        if (inv_ntt) {
            LL ni = inv(n, P);
            reverse(a + 1, a + n);
            for (i = 0; i < n; i++) a[i] = (a[i] * ni) % P;
        }
    }
};
const LL P = 2013265921, root = 31;
const int MAXN = 4194304;

```

```
NTT<P, root, MAXN> ntt;
```

3.9 質數表

```

// notPrime, 最大質因數, 質因數數量
int notPrime[MAXN], fac[MAXN], num[MAXN];
void getPrimes() {
    notPrime[1] = 1, fac[1] = num[1];
    for (int i = 2; i < MAXN; ++i) {
        if (notPrime[i]) continue;
        for (int j = i; j < MAXN; j += i) {
            if (i != j) notPrime[j] = 1;
            fac[j] = i, num[j]++;
        }
    }
}
vector<int> ret; // 質因數分解
void div(int x) {
    for (; x > 1; x /= fac[x]) ret.push_back(fac[x]);
}

```

3.10 歐拉函數

$\varphi(x)$ = 所有小於等於 x 且和 x 互質的數的數量

```

int _phi(int n) { // O(sqrtN)
    int res = n, a = n;
    for (int i = 2; i * i <= a; i++) {
        if (a % i == 0) {
            res = res / i * (i - 1);
            while (a % i == 0) a /= i;
        }
    }
    if (a > 1) res = res / a * (a - 1);
    return res;
}

```

```

int phi[MAXN]; // 建表 最大 1e7
void phi_table(int n) {
    phi[1] = 1;
    for (int i = 2; i <= n; ++i) {
        if (phi[i]) continue;
        for (int j = i; j <= n; j += i) {
            if (phi[j] == 0) phi[j] = j;
            phi[j] = phi[j] / i * (i - 1);
        }
    }
}

```

3.11 Miller-Rabin Algorithm

$$Magic = \begin{cases} \{2, 7, 61\}, & n < 4,759,123,141 \\ \{2, 13, 23, 1662803\}, & n < 1,122,004,669,633 \\ \{2, 3, 5, 7, 11, 13\}, & n < 3,474,749,660,383 \\ \{2, 325, 9375, 28178, \\ 450775, 9780504, 1795265022\} & n < 2^{64} \end{cases}$$

```

ll mul(ll x, ll y, ll mod) {
    ll ret = x * y - (ll)((long double)x / mod * y) * mod;
    return ret < 0 ? ret + mod : ret;
    // return x * y % mod; // for __int128
}
ll magic[] = {2, 7, 61}; // 這邊填入要用於測試的數列
ll S = 3; // 測試的數列數字的數量
bool witness(ll a, ll n, ll u, int t) {
    if (!a) return 0;
    ll x = power(a, u, n);
    for (int i = 0; i < t; i++) {
        ll nx = mul(x, x, n);
        if (nx == 1 && x != 1 && x != n - 1) return 1;
        x = nx;
    }
    return x != 1;
}
bool miller_rabin(ll n) {
    int s = S; // magic number size
    // iterate s times of witness on n
    if (n < 2) return false;
    if (!(n & 1)) return (n == 2);

    // 將 n - 1 寫成 u * (2 ^ t) 的形式
    ll u = n - 1;
    int t = 0;

```

```

while (!(u & 1)) u >>= 1, t++;

while (s--) {
    ll a = magic[s] % n;
    if (witness(a, n, u, t)) return false;
}
return true;
}

```

3.12 Pollard's Rho Algorithm

```

vector<ll> ret;
ll f(ll x, ll c, ll mod) {
    return (mul(x, x, mod) + c) % mod;
}
ll pollard_rho(ll n) {
    ll c = 1, x = 0, y = 0, p = 2, q, t = 0;
    while (t++ % 128 || gcd(p, n) == 1) {
        if (x == y) c++, y = f(x = 2, c, n);
        if (q = mul(p, abs(x - y), n)) p = q;
        x = f(x, c, n);
        y = f(f(y, c, n), c, n);
    }
    return gcd(p, n);
}
void fact(ll x) { // 透過遞迴找質數
    if (miller_rabin(x)) {
        ret.push_back(x);
        return;
    }
    ll f = pollard_rho(x);
    fact(f);
    fact(x / f);
}

```

3.13 高斯消去法

```

/**
 * mat[n][m] 為 n 個 m 元一次連列方程式的增廣矩陣
 * ans[m] 輸出 x_m 的解
 * pivot[n] 紀錄第 n 列的領導係數的位置
 * 若無解則回傳 false
 */
ll mat[MXN][MXN], ans[MXN];
int pivot[MXN];

/**
 * 回傳 true 代表有解，false 代表無解
 */
bool gaussian(int n, int m) {
    fill(pivot, pivot + m, -1);
    int row = 0;
    for (int col = 0; col < m && row < n; col++) {
        int sel = -1;
        for (int i = row; i < n; i++) { // 找這一欄係數不為 0 的列
            if (mat[i][col] != 0) {
                sel = i;
                break;
            }
        }
        if (sel == -1) // 全是自由變數，直接跳過處理下一個變數
            continue;
        // 把找到的那一列換到當前 row 的位置
        for (int j = col; j <= m; j++)
            swap(mat[sel][j], mat[row][j]);

        // 歸一化：把主元 (mat[row][col]) 變成 1
        for (int j = col; j <= m; j++)
            mat[row][j] = mat[row][j] / mat[row][col];

        for (int i = 0; i < n; i++) { // 消去
            if (i != row && mat[i][col] != 0) {
                ll factor = mat[i][col];
                for (int j = col; j <= m; j++)
                    mat[i][j] = mat[i][j] - factor * mat[row][j];
            }
        }
        // 紀錄這個變數是由第 row 列解出來的
        // 處理下一列
        pivot[col] = row++;
    }
}

```

```

}
// 檢查無解的情況
// 剩下的列 (從 row 到 n-1)，左邊係數應該全被消成 0 了
// 如果常數項 (mat[i][m]) 不為 0，表示 0 = k
// (k!=0)，即無解
for (int i = row; i < n; i++) {
    if (mat[i][m] != 0) return false;
}
for (int j = 0; j < m; j++) { // 填入答案
    if (pivot[j] != -1) // 對應的主元列的最右邊常數即為解
        ans[j] = mat[pivot[j]][m];
    else ans[j] = 0;
}
return true;
}

```

3.14 約瑟夫問題

```

int josephus(int n, int m) { // n 人每 m 次
    int ans = 0;
    for (int i = 1; i <= n; ++i) ans = (ans + m) % i;
    return ans;
}

```

3.15 莫比烏斯函數

```

vector<int> pri;
bool not_prime[MXN];
int mu[MXN];

void pre(int n) {
    mu[1] = 1;
    for (int i = 2; i <= n; ++i) {
        if (!not_prime[i]) {
            mu[i] = -1;
            pri.push_back(i);
        }
        for (int pri_j : pri) {
            if (i * pri_j > n) break;
            not_prime[i * pri_j] = true;
            if (i % pri_j == 0) {
                mu[i * pri_j] = 0;
                break;
            }
            mu[i * pri_j] = -mu[i];
        }
    }
}

```

3.16 離散對數

```

// gcd(a, n) = 1, 算 a^x == b (mod n)
ll BSGS(ll a, ll b, ll n) {
    a %= n, b %= n;
    if (b == 1) return 0; // a^0 = 1
    ll m = (ll)sqrt(n) + 1; // 保險一點
    unordered_map<ll, ll> mp;

    // 拆成 a^{i * m} == b * a^j (mod n)
    ll baby = b;
    for (ll j = 0; j <= m; j++) {
        mp[baby] = j;
        baby = baby * a % n;
    }

    ll step = fpow(a, m, n);
    ll giant = step;
    for (ll i = 1; i <= m; i++) { // 計算 (a^m)^i
        if (mp.count(giant)) {
            ll j = mp[giant], res = i * m - j;
            if (res >= 0) return res;
        }
        giant = giant * step % n;
    }

    return -1;
}

```

3.17 離散根號

```

void calcH(ll& t, ll& h, const ll p) {
    ll tmp = p - 1;
    for (t = 0; (tmp & 1) == 0; tmp /= 2) t++;
}

```

```

h = tmp;
}
// solve equation  $x^2 \bmod p = a$ 
bool solve(ll a, ll p, ll& x, ll& y) {
    if (p == 2) {
        x = y = 1;
        return true;
    }
    int p2 = p / 2, tmp = power(a, p2, p);
    if (tmp == p - 1) return false;
    if ((p + 1) % 4 == 0) {
        x = power(a, (p + 1) / 4, p);
        y = p - x;
        return true;
    } else {
        ll t, h, b, pb;
        calcH(t, h, p);
        if (t >= 2) {
            do {
                b = rand() % (p - 2) + 2;
            } while (power(b, p / 2, p) != p - 1);
            pb = power(b, h, p);
        }
        int s = power(a, h / 2, p);
        for (int step = 2; step <= t; step++) {
            int ss = (((ll)(s * s) % p) * a) % p;
            for (int i = 0; i < t - step; i++)
                ss = mul(ss, ss, p);
            if (ss + 1 == p) s = (s * pb) % p;
            pb = ((ll)pb * pb) % p;
        }
        x = ((ll)s * a) % p;
        y = p - x;
    }
    return true;
}

```

3.18 XOR Basis

```

const int LOGN = 31;
vector<int> find_basis(vector<int>& arr) {
    vector<int> basis(LOGN);
    for (int& x : arr) {
        for (int i = LOGN - 1; i >= 0; i--) {
            if ((x & (1 << i)) == 0) continue;
            if (!basis[i]) {
                basis[i] = x;
                break;
            } else x ^= basis[i];
        }
    }
    return basis;
}

bool check(int x) {
    for (int i = LOGN; i >= 0; i--) {
        if (x & (1 << i)) {
            if (!basis[i]) return false;
            x ^= basis[i];
        }
    }
    return true;
}

```

4 Graph

4.1 Topological Sort

```

vector<int> edge[MAXN], ans;
int deg[MAXN]; // in degree
void topo(int n) { // 1-base
    queue<int> que;
    for (int i = 1; i <= n; i++)
        if (!deg[i]) que.push(i);
    while (!que.empty()) {
        int u = que.front();
        que.pop();
        ans.push_back(u);
        // 結果存 ans
        for (int v : edge[u]) {
            deg[v]--;
            if (!deg[v]) que.push(v);
        }
    }
}

```

4.2 Tarjan's Algorithm for BCC

```

struct BccVertex {
    int n, nScc, step, dfn[MAXN], low[MAXN];
    vector<int> E[MAXN], sccv[MAXN];
    int top, stk[MAXN];
    void init(int _n) {
        n = _n;
        nScc = step = 0;
        for (int i = 0; i < n; i++)
            E[i].clear();
    }
    void addEdge(int u, int v) {
        E[u].PB(v);
        E[v].PB(u);
    }
    void DFS(int u, int f) {
        dfn[u] = low[u] = step++;
        stk[top++] = u;
        for (auto v : E[u]) {
            if (v == f) continue;
            if (dfn[v] == -1) {
                DFS(v, u);
                low[u] = min(low[u], low[v]);
                if (low[v] >= dfn[u]) {
                    int z;
                    sccv[nScc].clear();
                    do {
                        z = stk[--top];
                        sccv[nScc].PB(z);
                    } while (z != v);
                    sccv[nScc++].PB(u);
                } else low[u] = min(low[u], dfn[v]);
            }
        }
    }
    vector<vector<int>> solve() {
        vector<vector<int>> res;
        for (int i = 0; i < n; i++)
            dfn[i] = low[i] = -1;
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) {
                top = 0;
                DFS(i, i);
            }
        REP(i, nScc) res.PB(sccv[i]);
        return res;
    }
} graph;

```

4.3 Bellmen-Ford Algorithm

```

struct Edge {
    int from, to, weight;
} edge[MAXN];
int dis[MAXN];
void bellmen_ford(int n, int m) {
    dis[1] = 0; // 起點的距離一定是 0
    for (int j = 0; j < n - 1; j++) { // n 個節點要跑 n - 1 次
        for (int i = 0; i < m; i++) { // 對於所有邊都嘗試鬆弛
            if (dis[edge[i].to] >
                dis[edge[i].from] + edge[i].weight)
                dis[edge[i].to] =
                    dis[edge[i].from] + edge[i].weight;
        }
    }
}

```

4.4 Dijkstra's Algorithm

```

vector<pii> vec[N]; // vec[u] = {v, w}: u 為起點 v 為終點
// w 為路權
bool vis[N]; // 紀錄該節點是否走過
vector<int> dijkstra(int s) { // 起點
    vector<int> dis(N);
    for (int i = 0; i < N; i++) // 初始化
        dis[i] = INF; // 值要設為比可能的最短路徑權重還要大的值
    dis[s] = 0;
    priority_queue<pii, vector<pii>, greater<pii>>
        pq; // 以小到大批序
    // {w, v}: w 為路權 v 為節點
}

```

```

pq.push({dis[s], s});
while (pq.empty() == 0) {
    int u = pq.top().second;
    pq.pop();
    if (vis[u]) // 走過的不要再走
        continue;
    vis[u] = 1;
    for (auto i : vec[u]) {
        int v = i.first, w = i.second;
        if (dis[u] + w < dis[v]) { // 鬆弛
            dis[v] = dis[u] + w;
            pq.push({dis[v], v});
        }
    }
}
return dis;
}

```

4.5 Dinic's Algorithm for Maximum Flow

```

#define SZ(x) ((int)x.size())
#define PB push_back
struct Dinic {
    struct Edge {
        int v, f, re;
    };
    int n, s, t, level[MXN];
    vector<Edge> E[MXN];
    void init(int _n, int _s, int _t) {
        n = _n;
        s = _s;
        t = _t;
        for (int i = 0; i < n; i++) E[i].clear();
    }
    void add_edge(int u, int v, int f) {
        E[u].PB({v, f, SZ(E[v])});
        E[v].PB({u, 0, SZ(E[u]) - 1});
    }
    bool BFS() {
        for (int i = 0; i < n; i++) level[i] = -1;
        queue<int> que;
        que.push(s);
        level[s] = 0;
        while (!que.empty()) {
            int u = que.front();
            que.pop();
            for (auto it : E[u]) {
                if (it.f > 0 && level[it.v] == -1) {
                    level[it.v] = level[u] + 1;
                    que.push(it.v);
                }
            }
        }
        return level[t] != -1;
    }
    int DFS(int u, int nf) {
        if (u == t) return nf;
        int res = 0;
        for (auto &it : E[u]) {
            if (it.f > 0 && level[it.v] == level[u] + 1) {
                int tf = DFS(it.v, min(nf, it.f));
                res += tf;
                nf -= tf;
                it.f -= tf;
                E[it.v][it.re].f += tf;
                if (nf == 0) return res;
            }
        }
        if (!res) level[u] = -1;
        return res;
    }
    int flow(int res = 0) {
        while (BFS()) res += DFS(s, 2147483647);
        return res;
    }
} flow;
/*
    查找二分圖配對點:
    在 add_edge 時寫上 index[u].push_back({v,
    flow.E[u].size()});

```

在 flow 完之後寫上這個
 for (int i = 1; i <= n; i++) { // 遍歷所有節點

```

        for (pair<int, int> j : index[i]) {
            if (flow.E[i][j.second - 1].f == 0)
                cout << i << ' ' << j.first << '\n';
        }
    }
}
*/

```

4.6 Dominator Tree

```

#define REP(i, s, e) for (int i = (s); i <= (e); i++)
#define REPD(i, s, e) for (int i = (s); i >= (e); i--)
struct DominatorTree { // O(N)
    int n, s;
    vector<int> g[MAXN], pred[MAXN];
    vector<int> cov[MAXN];
    int dfn[MAXN], nfd[MAXN], ts;
    int par[MAXN]; // idom[u] s到u的最後一個必經點
    int sdom[MAXN], idom[MAXN];
    int mom[MAXN], mn[MAXN];
    inline bool cmp(int u, int v) {
        return dfn[u] < dfn[v];
    }
    int eval(int u) {
        if (mom[u] == u) return u;
        int res = eval(mom[u]);
        if (cmp(sdom[mn[mom[u]]], sdom[mn[u]]))
            mn[u] = mn[mom[u]];
        return mom[u] = res;
    }
    void init(int _n, int _s) {
        ts = 0;
        n = _n;
        s = _s;
        REP(i, 1, n) g[i].clear(), pred[i].clear();
    }
    void addEdge(int u, int v) {
        g[u].push_back(v);
        pred[v].push_back(u);
    }
    void dfs(int u) {
        ts++;
        dfn[u] = ts;
        nfd[ts] = u;
        for (int v : g[u])
            if (dfn[v] == 0) {
                par[v] = u;
                dfs(v);
            }
    }
    void build() {
        REP(i, 1, n) {
            dfn[i] = nfd[i] = 0;
            cov[i].clear();
            mom[i] = mn[i] = sdom[i] = i;
        }
        dfs(s);
        REPD(i, n, 2) {
            int u = nfd[i];
            if (u == 0) continue;
            for (int v : pred[u])
                if (dfn[v]) {
                    eval(v);
                    if (cmp(sdom[mn[v]], sdom[u]))
                        sdom[u] = sdom[mn[v]];
                }
            cov[sdom[u]].push_back(u);
            mom[u] = par[u];
            for (int w : cov[par[u]]) {
                eval(w);
                if (cmp(sdom[mn[w]], par[u])) idom[w] = mn[w];
                else idom[w] = par[u];
            }
            cov[par[u]].clear();
        }
        REP(i, 2, n) {
            int u = nfd[i];
            if (u == 0) continue;
            if (idom[u] != sdom[u]) idom[u] = idom[idom[u]];
        }
    }
} domT;

```

4.7 Edmonds Blossom Algorithm

```

namespace blossom {
#define d(x) (lab[x.u] + lab[x.v] - 2 * e[x.u][x.v].w)
#define all(x) x.begin(), x.end()
const ll N = 403 * 2;
const ll inf = 1e18;
struct Q {
    ll u, v, w;
} e[N][N];
vector<ll> p[N];
ll n, m = 0, id, h, t, lk[N], sl[N], st[N], f[N], b[N][N],
    s[N], ed[N], q[N], lab[N];
void upd(ll u, ll v) {
    if (!sl[v] || d(e[u][v]) < d(e[sl[v]][v])) sl[v] = u;
}
void ss(ll v) {
    sl[v] = 0;
    for (ll u = 1; u <= n; ++u)
        if (e[u][v].w > 0 && st[u] != v && !s[st[u]])
            upd(u, v);
}
void ins(ll u) {
    if (u <= n) q[++t] = u;
    else
        for (ll v : p[u]) ins(v);
}
void ch(ll u, ll w) {
    st[u] = w;
    if (u > n)
        for (ll v : p[u]) ch(v, w);
}
ll gr(ll u, ll v) {
    if ((v = find(all(p[u]), v) - p[u].begin()) & 1) {
        reverse(1 + all(p[u]));
        return (ll)p[u].size() - v;
    }
    return v;
}
void stm(ll u, ll v) {
    lk[u] = e[u][v].v;
    if (u <= n) return;
    Q w = e[u][v];
    ll x = b[u][w.u], y = gr(u, x);
    for (ll i = 0; i < y; ++i) stm(p[u][i], p[u][i ^ 1]);
    stm(x, v);
    rotate(p[u].begin(), y + all(p[u]));
}
void aug(ll u, ll v) {
    ll w = st[lk[u]];
    stm(u, v);
    if (!w) return;
    stm(w, st[f[w]]);
    aug(st[f[w]], w);
}
ll lca(ll u, ll v) {
    for (id++; u | v; swap(u, v)) {
        if (!u) continue;
        if (ed[u] == id) return u;
        ed[u] = id;
        if (u = st[lk[u]]) u = st[f[u]]; // =, not ==
    }
    return 0;
}
void add(ll u, ll a, ll v) {
    ll x = n + 1;
    while (x <= m && st[x]) ++x;
    if (x > m) ++m;
    lab[x] = s[x] = st[x] = 0;
    lk[x] = lk[a];
    p[x].clear();
    p[x].push_back(a);
#define op(q) \
    for (ll i = q, j = 0; i != a; i = st[f[j]]) \
    p[x].push_back(i), p[x].push_back(j = st[lk[i]]), \
    ins(j) // also not ==
    op(u);
    reverse(1 + all(p[x]));
    op(v);
    ch(x, x);
    for (ll i = 1; i <= m; ++i) e[x][i].w = e[i][x].w = 0;
    fill(b[x], b[x] + n + 1, 0);
    for (ll u : p[x]) {
        for (ll v = 1; v <= m; ++v)
            if (!e[x][v].w || d(e[u][v]) < d(e[x][v]))
                e[x][v] = e[u][v], e[v][x] = e[v][u];
    }
}

```

```

    for (ll v = 1; v <= n; ++v)
        if (b[u][v]) b[x][v] = u;
    }
    ss(x);
}
void ex(ll u) {
    for (ll x : p[u]) ch(x, x);
    ll a = b[u][e[u][f[u]].u], r = gr(u, a);
    for (ll i = 0; i < r; i += 2) {
        ll x = p[u][i], y = p[u][i + 1];
        f[x] = e[y][x].u;
        s[x] = 1;
        s[y] = 0;
        sl[x] = 0;
        ss(y);
        ins(y);
    }
    s[a] = 1;
    f[a] = f[u];
    for (ll i = r + 1; i < p[u].size(); ++i)
        s[p[u][i]] = -1, ss(p[u][i]);
    st[u] = 0;
}
bool on(const Q &e) {
    ll u = st[e.u], v = st[e.v], a;
    if (s[v] == -1) {
        f[v] = e.u, s[v] = 1, a = st[lk[v]];
        sl[v] = sl[a] = s[a] = 0, ins(a);
    } else if (!s[v]) {
        a = lca(u, v);
        if (!a) return aug(u, v), aug(v, u), 1;
        else add(u, a, v);
    }
    return 0;
}
bool bfs() {
    fill(s, s + m + 1, -1);
    fill(sl, sl + m + 1, 0); // s is filled with -1
    h = 1, t = 0;
    for (ll i = 1; i <= m; ++i)
        if (st[i] == i && !lk[i]) f[i] = s[i] = 0, ins(i);
    if (h > t) return 0;
    while (1) {
        while (h <= t) {
            ll u = q[h++];
            if (s[st[u]] != 1) {
                for (ll v = 1; v <= n; ++v)
                    if (e[u][v].w > 0 && st[u] != st[v]) {
                        if (d(e[u][v])) upd(u, st[v]);
                        else if (on(e[u][v])) return 1;
                    }
            }
        }
        ll x = inf;
        for (ll i = n + 1; i <= m; ++i)
            if (st[i] == i && s[i] == 1) x = min(x, lab[i] / 2);
        for (ll i = 1; i <= m; ++i)
            if (st[i] == i && sl[i] && s[i] != 1)
                x = min(x, d(e[sl[i]][i]) >> s[i] + 1);
        for (ll i = 1; i <= n; ++i)
            if (~s[st[i]])
                if ((lab[i] += (s[st[i]] * 2 - 1) * x) <= 0)
                    return 0;
        for (ll i = n + 1; i <= m; ++i)
            if (st[i] == i && ~s[st[i]])
                lab[i] += (2 - 4 * s[st[i]]) * x;
        h = 1, t = 0;
        for (ll i = 1; i <= m; ++i)
            if (st[i] == i && sl[i] && st[sl[i]] != i &&
                !d(e[sl[i]][i]) && on(e[sl[i]][i]))
                return 1;
        for (ll i = n + 1; i <= m; ++i)
            if (st[i] == i && s[i] == 1 && !lab[i]) ex(i);
    }
}
pair<ll, vector<array<ll, 2>>> solve(
    ll N, vector<array<ll, 3>>> edges) {
    for (auto &edge : edges) ++edge[0], ++edge[1];
    fill(ed, ed + m + 1, 0);
    fill(lk, lk + m + 1, 0);
    n = m = N;
    id = 0;
    iota(st + 1, st + n + 1, 1);
}

```

```

ll wm = 0, weight = 0;
for (ll i = 1; i <= n; ++i)
    for (ll j = 1; j <= n; ++j) e[i][j] = {i, j, 0};
for (auto edge : edges) {
    int u = edge[0], v = edge[1];
    ll w = edge[2];
    wm = max(wm,
        e[v][u].w = e[u][v].w = max(e[u][v].w, w));
}
for (ll i = 1; i <= n; ++i) p[i].clear();
for (ll i = 1; i <= n; ++i)
    for (ll j = 1; j <= n; ++j) b[i][j] = i == j ? 1 : 0;
fill_n(lab + 1, n, wm);
while (bfs()) {}
vector<array<ll, 2>> matching;
for (ll i = 1; i <= n; ++i)
    if (i < lk[i])
        weight += e[i][lk[i]].w,
        matching.push_back({i - 1, lk[i] - 1});
return {weight, matching};
}
// namespace blossom

```

4.8 Floyd-Warshall Algorithm

```

int dis[N][N];
void init(int n) {
    for (int i = 0; i <= n; i++) { // 初始化
        for (int j = 0; j <= n; j++)
            dis[i][j] = INF;
        dis[i][i] = 0;
    }
}
void floyd_warshall(int n) {
    for (int k = 0; k < n; k++) { // 窮舉中繼點 k
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) { // 窮舉點對 (i, j)
                dis[i][j] = min(dis[i][j], dis[i][k] + dis[k][j]);
            }
        }
    }
}

```

4.9 Maximum Clique

```

#define N 111 // 80 ~ 100 左右
struct MaxClique { // 0-base
    typedef bitset<N> Int;
    Int linkto[N], v[N];
    int n;
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; i++) {
            linkto[i].reset();
            v[i].reset();
        }
    }
    void addEdge(int a, int b) { v[a][b] = v[b][a] = 1; }
    int popcount(const Int& val) { return val.count(); }
    int lowbit(const Int& val) { return val._Find_first(); }
    int ans, stk[N];
    int id[N], di[N], deg[N];
    Int cans;
    void maxclique(int elem_num, Int candi) {
        if (elem_num > ans) {
            ans = elem_num;
            cans.reset();
            for (int i = 0; i < elem_num; i++)
                cans[id[stk[i]]] = 1;
        }
        int potential = elem_num + popcount(candi);
        if (potential <= ans) return;
        int pivot = lowbit(candi);
        Int smaller_candi = candi & (~linkto[pivot]);
        while (smaller_candi.count() && potential > ans) {
            int next = lowbit(smaller_candi);
            candi[next] = !candi[next];
            smaller_candi[next] = !smaller_candi[next];
            potential--;
            if (next == pivot ||
                (smaller_candi & linkto[next]).count()) {
                stk[elem_num] = next;
                maxclique(elem_num + 1, candi & linkto[next]);
            }
        }
    }
}

```

```

}
}
int solve() {
    for (int i = 0; i < n; i++) {
        id[i] = i;
        deg[i] = v[i].count();
    }
    sort(id, id + n, [&](int id1, int id2) {
        return deg[id1] > deg[id2];
    });
    for (int i = 0; i < n; i++)
        di[id[i]] = i;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            if (v[i][j]) linkto[di[i]][di[j]] = 1;
    Int cand;
    cand.reset();
    for (int i = 0; i < n; i++)
        cand[i] = 1;
    ans = 1;
    cans.reset();
    cans[0] = 1;
    maxclique(0, cand);
    return ans;
}
} solver;

```

4.10 Kosaraju's Algorithm for SCC

```

struct Scc { // 新的 DAG 可能會有多重邊
#define PB push_back
    int n, nScc, vst[MXN], bln[MXN];
    vector<int> E[MXN], rE[MXN], vec, dag[MXN];
    set<int> vis[MXN];
    void init(int _n) {
        n = _n;
        for (int i = 0; i <= n; i++) {
            E[i].clear(), rE[i].clear();
            dag[i].clear(), vis[i].clear();
            bln[i] = -1;
        }
    }
    void addEdge(int u, int v) {
        E[u].PB(v);
        rE[v].PB(u);
    }
    void DFS(int u) {
        vst[u] = 1;
        for (auto v : E[u])
            if (!vst[v]) DFS(v);
        vec.PB(u);
    }
    void rDFS(int u) {
        vst[u] = 1;
        bln[u] = nScc; // 編號是 0-base
        for (auto v : rE[u])
            if (!vst[v]) rDFS(v);
    }
    void build_new_dag() {
        for (int i = 0; i < n; i++) {
            for (int &j : E[i]) {
                if (bln[i] != bln[j] &&
                    vis[bln[i]].count(bln[j]) == 0) {
                    dag[bln[i]].push_back(bln[j]);
                    vis[bln[i]].insert(bln[j]);
                }
            }
        }
    }
    void solve() {
        nScc = 0;
        vec.clear();
        fill(vst, vst + n + 1, 0);
        for (int i = 0; i < n; i++)
            if (!vst[i]) DFS(i);
        reverse(vec.begin(), vec.end());
        fill(vst, vst + n + 1, 0);
        for (auto v : vec) {
            if (!vst[v]) {
                rDFS(v);
                nScc++;
            }
        }
        build_new_dag();
    }
}

```

```

}
};

```

4.11 Kruskal's Algorithm for MST

```

void kruskal() {
    sort(graph, graph + m); // 將邊照大小排序
    int ans = 0;

    for (int i = 0; i < m; i++) { //>:
        if (find(graph[i].u) != find(graph[i].v)) {
            // 如果兩點未聯通
            merge(graph[i].u, graph[i].v);
            // 將兩點設成同一個集合
            ans += graph[i].w; // 權重加進答案
            if (sz[find(graph[i].u)] == n)
                break; // 當並查集大小等價於樹內點的數量
        }
    }
}

```

4.12 Kuhn-Munkre Algorithm

```

struct KM { // max weight, for min negate the weights
    int n, mx[MXN], my[MXN], pa[MXN];
    ll g[MXN][MXN], lx[MXN], ly[MXN], sy[MXN];
    bool vx[MXN], vy[MXN];
    void init(int _n) { // 1-based
        n = _n; // 初始化
        for (int i = 1; i <= n; i++)
            fill(g[i], g[i] + n + 1, 0);
    }
    void addEdge(int x, int y, ll w) { // 加邊
        g[x][y] = w;
    }
    void augment(int y) {
        for (int x, z; y; y = z)
            x = pa[y], z = mx[x], my[y] = x, mx[x] = y;
    }
    void bfs(int st) {
        for (int i = 1; i <= n; ++i)
            sy[i] = INF, vx[i] = vy[i] = 0;
        queue<int> q;
        q.push(st);
        for (;;) {
            while (q.size()) {
                int x = q.front();
                q.pop();
                vx[x] = 1;
                for (int y = 1; y <= n; ++y)
                    if (!vy[y]) {
                        ll t = lx[x] + ly[y] - g[x][y];
                        if (t == 0) {
                            pa[y] = x;
                            if (!my[y]) {
                                augment(y);
                                return;
                            }
                        }
                        vy[y] = 1, q.push(my[y]);
                    } else if (sy[y] > t) pa[y] = x, sy[y] = t;
            }
            ll cut = INF;
            for (int y = 1; y <= n; ++y)
                if (!vy[y] && cut > sy[y]) cut = sy[y];
            for (int j = 1; j <= n; ++j) {
                if (vx[j]) lx[j] -= cut;
                if (vy[j]) ly[j] += cut;
                else sy[j] -= cut;
            }
            for (int y = 1; y <= n; ++y)
                if (!vy[y] && sy[y] == 0) {
                    if (!my[y]) {
                        augment(y);
                        return;
                    }
                    vy[y] = 1, q.push(my[y]);
                }
        }
    }
    ll solve() { // 解決
        fill(mx, mx + n + 1, 0);
        fill(my, my + n + 1, 0);
    }
}

```

```

fill(ly, ly + n + 1, 0);
fill(lx, lx + n + 1, -INF);
for (int x = 1; x <= n; ++x)
    for (int y = 1; y <= n; ++y)
        lx[x] = max(lx[x], g[x][y]);
for (int x = 1; x <= n; ++x)
    bfs(x);
ll ans = 0;
for (int y = 1; y <= n; ++y)
    ans += g[my[y]][y]; // 答案存在 my[]
return ans;
}
} graph; // 找二分圖最大權完美匹配

```

4.13 Max-flow-min-cost

```

struct zkwflow {
    static const int maxN = 100000;
    struct Edge {
        int v, f, re;
        ll w;
    };
    int n, s, t, ptr[maxN];
    bool vis[maxN];
    ll dis[maxN];
    vector<Edge> E[maxN];
    void init(int _n, int _s, int _t) {
        n = _n, s = _s, t = _t;
        for (int i = 0; i < n; i++)
            E[i].clear();
    }
    void add_edge(int u, int v, int f, ll w) {
        E[u].push_back({v, f, (int)E[v].size(), w});
        E[v].push_back({u, 0, (int)E[u].size() - 1, -w});
    }
    bool SPFA() {
        fill_n(dis, n, LLONG_MAX);
        fill_n(vis, n, false);
        queue<int> q;
        q.push(s);
        dis[s] = 0;
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            vis[u] = false;
            for (auto &it : E[u]) {
                if (it.f > 0 && dis[it.v] > dis[u] + it.w) {
                    dis[it.v] = dis[u] + it.w;
                    if (!vis[it.v]) {
                        vis[it.v] = true;
                        q.push(it.v);
                    }
                }
            }
        }
        return dis[t] != LLONG_MAX;
    }
    int DFS(int u, int nf) {
        if (u == t) return nf;
        int res = 0;
        vis[u] = true;
        for (int &i = ptr[u]; i < (int)E[u].size(); i++) {
            auto &it = E[u][i];
            if (it.f > 0 && dis[it.v] == dis[u] + it.w &&
                !vis[it.v]) {
                int tf = DFS(it.v, min(nf, it.f));
                res += tf, nf -= tf, it.f -= tf;
                E[it.v][it.re].f += tf;
                if (nf == 0) {
                    vis[u] = false;
                    break;
                }
            }
        }
        return res;
    }
    pair<int, ll> flow() {
        int flow = 0;
        ll cost = 0;
        while (SPFA()) {
            fill_n(ptr, n, 0);
            int f = DFS(s, INT_MAX);
            flow += f;
            cost += dis[t] * f;
        }
    }
}

```

```

    }
    return {flow, cost};
} // reset: do nothing
} flow;

```

4.14 Kuhn's Algorithm

```

vector<int> g[MXN];
bool vis[MXN];
int S[MXN]; // S 為紀錄這個點與誰匹配
int n, ans; // n: 左集合數量, ans: 紀錄答案

```

```

bool dfs(int u) { // 找最大匹配
    for (int &v : g[u]) {
        if (!vis[v]) {
            vis[v] = true;
            if (S[v] == -1 || dfs(S[v])) {
                S[v] = u;
                return true;
            }
        }
    }
    return false;
}

void solve() { // 記得每次使用需清空 vis
    memset(S, -1, sizeof(S));
    ans = 0;
    for (int i = 1; i <= n; i++) { // 跑左集合
        memset(vis, 0, sizeof(vis));
        if (dfs(i)) ans++;
    }
}

```

4.15 Flow Methods

【二分圖定理】($M = \text{Max Matching}$)

- $\text{Min Vertex Cover (MVC)} = M$
- $\text{Max Independent Set (MIS)} = |V| - M$
- $\text{Min Edge Cover} = |V| - M$

注意：LP 對偶提醒：Min Cut = Max Flow，遇到 Min/Max 轉換卡關時，寫出標準型轉 Dual。

【找二分圖 MVC & MIS 成員】

跑完 Dinic 後，在殘留網路上從 Source 走「Cap > 0」的邊進行 DFS/BFS：

- MVC = (左側未 visit 的點) U (右側有 visit 的點)
- MIS = (左側有 visit 的點) U (右側未 visit 的點) (即為 MVC 的補集)

【最大密度子圖】密度 $D = (\sum W_e + \sum W_v) / |V|$
對答案 D 進行二分搜。取極大常數 $S = \sum(\text{所有權重})$ 或 inf ，確保邊權非負。

建圖規則 (Cap)：

1. $S \rightarrow \text{點 } v : S$
 2. 原圖無向邊 (u, v, w) ：建雙向邊 $u \rightarrow v$ 及 $v \rightarrow u$ ，Cap = w
 3. 點 $v \rightarrow T : S + 2 * D - \text{deg}[v] - 2 * W_v$
(註：deg[v] 為與 v 相連的「邊權重」總和)
判定與找成員：
- 若 $\text{Max Flow} < S * |V|$ ，代表存在密度 > D 的子圖 (D 可繼續往上搜)。
 - 找子圖成員：跑完後從 S 走「Cap > 0」的邊，能到達的點集即為所求。

【PSP】

全部可能的收益 - Min Cut = 最大收益
做完 B 實驗才能做 A 實驗，則 $A \rightarrow B$

5 Tree Algorithms

5.1 時間戳記 & 倍增表 (初始化)

```

const int LG_MXN = __lg(MXN) + 2;
vector<int> edge[MXN]; // i 有哪些小孩
int timing = 1, lgN, n;
int in[MXN], out[MXN], anc[MXN][LG_MXN], depth[MXN];
void dfs_init(int u, int f) { // 0 倍祖先、時間戳記
    in[u] = timing++; // 這時進入 u
    anc[u][0] = f; // now 的 0 倍祖先是 f
    for (int v : edge[u]) {
        if (v == f) continue;
        depth[v] = depth[u] + 1;
    }
}

```

```

    dfs_init(v, u);
}
out[u] = timing++; // 這時離開 u
}

void build_table(int n) { // 初始化
    lgN = __lg(n) + 1;
    dfs_init(1, 0);
    for (int i = 1; i <= lgN; i++) // 建立倍增表
        for (int now = 1; now <= n; now++)
            anc[now][i] = anc[anc[now][i - 1]][i - 1];
}

```

5.2 LCA

```

bool is_ancestor(int u, int v) { // 判斷 u 是否 v 是祖先
    return (in[u] < in[v] && out[v] < out[u]);
}

int getLCA(int x, int y) { // 找最近共同祖先
    if (is_ancestor(x, y)) // 如果 x 為 y 的祖先
        return x; // 則 LCA 為 y
    if (is_ancestor(y, x)) // 如果 y 為 x 的祖先
        return y; // LCA 為 y
    for (int i = lgN; i >= 0; i--) // 逼近 LCA
        if (!is_ancestor(anc[x][i], y) && anc[x][i])
            x = anc[x][i];
    return anc[x][0]; // 回傳此點的父節點即為答案
}

```

5.3 Euler Tour for LCA

```

#include <bits/stdc++.h>
using namespace std;
const int N = 1e5 + 5;
vector<int> tour, edge[N];
int timing = 0;
int in[N], out[N];
void dfs(int u, int fa) {
    tour.push_back(u); // 進入的時間點
    in[u] = timing++;
    for (auto v : edge[u]) {
        if (v == fa) continue;
        dfs(v, u);
        tour.push_back(u); // 走完其中一個孩子 再走回自己
    }
    out[u] = timing++;
}

void build(vector<int>& tour) { // 把 tour 丟進線段樹初始化
    seg.build(tour); // 除了 tour，還要建每個點的深度
}

int query(int x, int y) {
    if (is_ancestor(x, y)) return x; // LCA(x, y) = x
    if (is_ancestor(y, x)) return y; // LCA(x, y) = y;
    // 如果「y的區間」在「x的區間」的左邊，就交換
    if (out[y] < in[x]) swap(x, y);
    // 取得區間內哪個位置的深度是最低的
    int index = seg.query(out[x], in[y]);
    return tour[index];
}

```

5.4 樹上差分

```

int tag[N], f[N], ans[N], root;
void add(int x, int y) {
    int lca = getLCA(x, y);
    tag[x]++;
    tag[y]++;
    tag[lca]--;
    if (lca != root) tag[f[lca]]--;
}

int dfs_add(int now, int fa) {
    int diff = tag[now];
    for (auto nxt : edge[now]) {
        if (nxt == fa) continue;
        diff += dfs_add(nxt, now);
    }
    return ans[now] = diff;
}

```

5.5 DSU on Tree

```

int find(int x) {
    if (f[x] == x) return x;
}

```

```

    else return f[x] = find(f[x]);
}
void merge(int x, int y) {
    x = find(x), y = find(y);
    if (sz[x] < sz[y]) swap(x, y);
    sz[x] += sz[y];
    f[y] = x;
}
void init() {
    // 初始化一開始大小都為1 (每群一開始都是獨立一個)
    for (int i = 1; i <= n; i++) sz[i] = 1;
    for (int i = 1; i <= n; i++) f[i] = i;
}

```

5.6 樹鏈剖分

```

#define REP(i, s, e) for (int i = (s); i <= (e); i++)
#define REPD(i, s, e) for (int i = (s); i >= (e); i--)
const int MAXN = 100010;
const int LOG = 19;
struct HLD {
    int n;
    vector<int> g[MAXN];
    int sz[MAXN], dep[MAXN];
    int ts, tid[MAXN], tdi[MAXN], tl[MAXN], tr[MAXN];
    // ts : timestamp, useless after yutruli
    // tid[ u ] : pos. of node u in the seq.
    // tdi[ i ] : node at pos i of the seq.
    // tl, tr[ u ] : subtree interval in the seq. of node
    // u
    int prt[MAXN][LOG], head[MAXN];
    // head[ u ] : head of the chain contains u
    void dfsz(int u, int p) {
        dep[u] = dep[p] + 1;
        prt[u][0] = p;
        sz[u] = 1;
        head[u] = u;
        for (int& v : g[u])
            if (v != p) {
                dep[v] = dep[u] + 1;
                dfsz(v, u);
                sz[u] += sz[v];
            }
    }
    void dfs1(int u) {
        ts++;
        tid[u] = tl[u] = tr[u] = ts;
        tdi[tid[u]] = u;
        sort(ALL(g[u]),
            [&](int a, int b) { return sz[a] > sz[b]; });
        bool flag = 1;
        for (int& v : g[u])
            if (v != prt[u][0]) {
                if (flag) head[v] = head[u], flag = 0;
                dfs1(v);
                tr[u] = tr[v];
            }
    }
    inline int lca(int a, int b) {
        if (dep[a] > dep[b]) swap(a, b);
        int diff = dep[b] - dep[a];
        REPD(k, LOG - 1, 0) if (diff & (1 << k)) {
            b = prt[b][k];
        }
        if (a == b) return a;
        REPD(k, LOG - 1, 0) if (prt[a][k] != prt[b][k]) {
            a = prt[a][k];
            b = prt[b][k];
        }
        return prt[a][0];
    }
    void init(int _n) {
        n = _n;
        REP(i, 1, n) g[i].clear();
    }
    void addEdge(int u, int v) {
        g[u].push_back(v);
        g[v].push_back(u);
    }
    void yutruli() { // build function
        dfsz(1, 0);
        ts = 0;
        dfs1(1);
        REP(k, 1, LOG - 1)

```

```

        REP(i, 1, n) prt[i][k] = prt[prt[i][k - 1]][k - 1];
    }
    vector<PII> getPath(int u, int v) {
        vector<PII> res;
        while (tid[u] < tid[head[v]]) {
            res.push_back(PII(tid[head[v]], tid[v]));
            v = prt[head[v]][0];
        }
        res.push_back(PII(tid[u], tid[v]));
        reverse(ALL(res));
        return res;
    }
    /* res : list of intervals from u to v
     * u must be ancestor of v
     * usage :
     * vector< PII >& path = tree.getPath( u , v )
     * for( PII tp : path ) {
     *     int l , r; tie( l , r ) = tp;
     *     upd( l , r );
     *     uu = tree.tdi[ l ], vv = tree.tdi[ r ];
     *     uu ~> vv is a heavy path on tree
     * }
     */
    int query(int x, int y) {
        int lca = getlca(x, y);
        int ans = 0;

        // x -> lca 點權總和
        while (top[lca] != top[x]) {
            ans += seg.query(dfn[top[x]], dfn[x]);
            x = par[top[x]];
        }
        ans += seg.query(dfn[lca], dfn[x]);

        // y -> lca 點權總和
        while (top[lca] != top[y]) {
            ans += seg.query(dfn[top[y]], dfn[y]);
            y = par[top[y]];
        }
        ans += seg.query(dfn[lca], dfn[y]);

        ans -= seg.query(
            dfn[lca],
            dfn[lca]); // lca被重複算到一次，要扣掉
    }
} tree;

```

5.7 虛樹

```

vector<int> virTree(vector<int> ver) {
    sort(ver.begin(), ver.end(), cmp); // 用dfn排序
    vector<int> res(ver.begin(), ver.end());
    for (int i = 1; i < ver.size(); i++) {
        res.push_back(
            lca(ver[i - 1], ver[i])); // 把LCA丟進虛樹內
    }
    sort(res.begin(), res.end(), cmp); // 在用dfn排序
    res.erase(
        unique(res.begin(), res.end()),
        res.end()); // 可能有重複的點，需要去掉重複的
    return res;
}
int count_answer(vector<int> virTree) {
    sort(virTree.begin(), virTree.end(), cmp);
    int ans = 0;
    for (int i = 1; i < virTree.size(); i++) {
        ans += query(lca(virTree[i - 1], virTree[i]),
            virTree[i]);
    }
    return ans;
}

```

5.8 樹重心

```

int cent; // 這就是重心
void dfs(int u, int pre) {
    sz[u] = 1;
    int mx = 0;
    for (int& v : g[u]) {
        if (v == pre) continue;
        dfs(v, u);
        sz[u] += sz[v];
        mx = max(mx, sz[v]);
    }
}

```

```

}
mx = max(mx, n - sz[u]);
if (mx <= n / 2) cent = u;
}

```

5.9 重心剖分

```

/* 以上略 */
const int MXN = 2e5 + 5;
vector<int> g[MXN];
int n, k, mx_dep;
ll ans = 0;
bool vis[MXN];
int sz[MXN], cnt[MXN] = {1};

// 計算以 u 為根（忽略已移除的重心與父邊）的子樹大小
int get_sz(int u, int pre) { // 這個先預處理
    sz[u] = 1;
    for (int& v : g[u]) {
        // 不跨越父親、也不進入已「移除當過重心」的節點
        if (v == pre || vis[v]) continue;
        sz[u] += get_sz(v, u);
    }
    return sz[u];
}

// 在包含 u 的當前子樹內找重心：
// 從 u 出發，若存在子樹大小 >= desired (約 n/2)
// 就往那個子樹走
// 直到找不到更大的子樹為止，回傳當前節點作為重心
int get_cent(int u, int pre, int desired) {
    for (int& v : g[u]) {
        if (v != pre && sz[v] >= desired && !vis[v])
            return get_cent(v, u, desired);
    }
    return u;
}

void cal_cnt(int u, int pre, bool filling, int dep = 1) {
    if (dep > k) return;
    mx_dep = max(
        mx_dep,
        dep); // 記錄本層最大深度，供稍後清空 cnt[] 使用
    if (filling)
        cnt[dep]++; // 填表：
        // 把這個子樹的距離分布，併入「已處理過的
        // 子樹」
    else
        ans += cnt[k - dep]; // 查詢：找另一側距離是 (k -
        // dep) 的節點數，配對成長度 k
    for (int& v : g[u]) {
        if (!vis[v] && pre != v)
            cal_cnt(v, u, filling, dep + 1);
    }
}

void decomp(int u = 1) {
    // 找重心（若 total 偶數，可能落在任一個重心）
    int cent = get_cent(u, 0, get_sz(u, 0) / 2);

    vis[cent] = true; // 移除重心，切割成多個子樹
    mx_dep = 0; // 重置本層使用過的最大深度

    // 針對每個鄰接子樹：先查詢、後填表
    // (避免同一子樹內互相配對)
    for (int& v : g[cent]) {
        if (vis[v]) continue;
        cal_cnt(v, cent,
            false); // 查詢：累加所有「經過重心」的長度=
            // k 的路徑
        cal_cnt(
            v, cent,
            true); // 填表：把這個子樹的距離分布併入 cnt[]
    }

    // 重置 cnt[]：保留 cnt[0]=1
    // (代表僅重心)，清掉本層曾經動用過的深度範圍 註：
    // 此實作選擇在「離開當前重心」時做清理，確保下一層剛進
    // 來時
    // cnt 狀態就是正確的
    cnt[0] = 1;

```

```

for (int i = 1; i <= mx_dep; i++) cnt[i] = 0;

// 遞迴處理每個子樹
// (此時重心已被標記，不會跨越到其他分支)
for (int& v : g[cent]) {
    if (!vis[v]) decomp(v);
}
}

```

5.10 Tree Hash

```

map<vector<int>, int> id;
int dfs(int x, int f) {
    vector<int> sub;
    for (int v : g[x]) {
        if (v != f) sub.push_back(dfs(v, x));
    }
    sort(sub.begin(), sub.end());
    if (!id.count(sub)) id[sub] = id.size();
    return id[sub];
}

```

6 String Algorithms

6.1 Suffix Array (Binary Lifting)

```

struct SA {
    int n;
    vector<int> rk, oldrk, sa, h;
    string s;
    SA(string& t): s(t), n(t.size()), rk(n<<1), oldrk(n
        <<1), sa(n), h(n) {
        for(int i = 0; i < n; i++) rk[i] = s[i] - 'a' + 1, sa[
            i] = i;

        if (n == 1) {
            rk[0] = 1;
            return;
        }

        for(int w = 1; w < n; w<=1) {
            sort(sa.begin(), sa.end(), [&](int x, int y){
                return (rk[x]==rk[y]) ? (rk[x+w]<rk[y+w])
                    : (rk[x]<rk[y]);
            });
            oldrk = rk;
            rk[sa[0]] = 1;
            for(int i = 1, p = 1; i < n; i++) {
                if(oldrk[sa[i]]==oldrk[sa[i-1]] && oldrk[
                    sa[i]+w]==oldrk[sa[i-1]+w]) rk[sa[i]]
                    = p;
                else rk[sa[i]] = ++p;
            }

            for(int i = 0, k = 0; i < n; i++) {
                if(rk[i] == 1) continue;
                if(k) k--;
                int j = sa[rk[i]-2];
                while(i+k < n && j+k < n && s[i+k] == s[j+k])
                    k++;
                h[rk[i]-1] = k;
            }
        }

        void find(string p) { // binary search
            int l = 0, r = n, m = p.size();
            if (m > n) return;
            for (int i = 0; i < m; i++) {
                auto cmp = [&](int lhs, int rhs) {
                    return s[lhs + i] < rhs;
                };
                auto nl =
                    lower_bound(sa + 1, sa + r, p[i], cmp) - sa;
                auto nr =
                    lower_bound(sa + 1, sa + r, p[i] + 1, cmp) - sa;
                l = nl, r = nr;
                if (l >= r) break;
            }
        }
        cout << (r > 1 ? "YES" : "NO") << "\n";
    }
};

```

6.2 Suffix Array (SAIS)

```
// 此為後綴陣列模板
// 用法:
// 把要處理的字串轉成整數陣列傳入suffix_array(), 會得到SA
// 與H維結果
const int N = 500010; // 有可能要在這就寫原本字串的兩倍長
struct SA {
#define REP(i, n) for (int i = 0; i < int(n); i++)
#define REP1(i, a, b) for (int i = (a); i <= int(b); i++)
    bool _t[N * 2];
    int _s[N * 2], _sa[N * 2], _c[N * 2], x[N], _p[N],
        _q[N * 2], hei[N], r[N];
    int operator[](int i) { return _sa[i]; }
    void build(int* s, int n, int m) {
        memcpy(_s, s, sizeof(int) * n);
        sais(_s, _sa, _p, _q, _t, _c, n, m);
        mkhei(n);
    }
    void mkhei(int n) {
        REP(i, n) r[_sa[i]] = i;
        hei[0] = 0;
        REP(i, n) if (r[i]) {
            int ans = i > 0 ? max(hei[r[i] - 1] - 1, 0) : 0;
            while (_s[i + ans] == _s[_sa[r[i] - 1] + ans])
                ans++;
            hei[r[i]] = ans;
        }
    }
    void sais(int* s, int* sa, int* p, int* q, bool* t,
        int* c, int n, int z) {
        bool uniq = t[n - 1] = true, neq;
        int nn = 0, nmzx = -1, *nsa = sa + n, *ns = s + n,
            lst = -1;
#define MS0(x, n) memset((x), 0, n * sizeof(*(x)))
#define MAGIC(XD) \
        MS0(sa, n); \
        memcpy(x, c, sizeof(int) * z); \
        XD; \
        memcpy(x + 1, c, sizeof(int) * (z - 1)); \
        REP(i, n) \
        if (sa[i] && !t[sa[i] - 1]) \
            sa[x[s[sa[i] - 1]]++] = sa[i] - 1; \
        memcpy(x, c, sizeof(int) * z); \
        for (int i = n - 1; i >= 0; i--) \
            if (sa[i] && t[sa[i] - 1]) \
                sa[--x[s[sa[i] - 1]]] = sa[i] - 1; \
        MS0(c, z); \
        REP(i, n) uniq &= ++c[s[i]] < 2; \
        REP(i, z - 1) c[i + 1] += c[i]; \
        if (uniq) { \
            REP(i, n) sa[--c[s[i]]] = i; \
            return; \
        } \
        for (int i = n - 2; i >= 0; i--) \
            t[i] = \
                (s[i] == s[i + 1] ? t[i + 1] : s[i] < s[i + 1]); \
        MAGIC(REP1(i, 1, n - 1) if (t[i] && !t[i - 1]) \
            sa[--x[s[i]]] = p[q[i] = nn++] = i); \
        REP(i, n) if (sa[i] && t[sa[i]] && !t[sa[i] - 1]) { \
            neq = lst < 0 || memcmp(s + sa[i], s + lst, \
                (p[q[sa[i]] + 1] - sa[i]) * \
                sizeof(int)); \
            ns[q[lst = sa[i]]] = nmzx += neq; \
        } \
        sais(ns, nsa, p + nn, q + n, t + n, c + z, nn, \
            nmzx + 1); \
        MAGIC(for (int i = nn - 1; i >= 0; i--) \
            sa[--x[s[p[nsa[i]]]]] = p[nsa[i]]); \
    }
} sa;
int H[N], SA[N]; // SA: 後綴陣列的Length
// H[i]: SA[i]與SA[i-1]的共同子字串長
void suffix_array(int* ip, int len) {
    // should padding a zero in the back
    // ip is int array, len is array length
    // ip[0..n-1] != 0, and ip[len] = 0
    ip[len++] = 0;
    sa.build(ip, len, 128);
    for (int i = 0; i < len; i++) {
        H[i] = sa.hei[i + 1];
        SA[i] = sa._sa[i + 1];
    }
}
```

```
// resulting height, sa array \in [0, len)
}
```

6.3 AC Automata

```
struct ACautomata {
    struct Node {
        int cnt, i;
        Node *go[26], *fail, *dic;
        Node() {
            cnt = 0;
            fail = 0;
            dic = 0;
            i = 0;
            memset(go, 0, sizeof(go));
        }
    } pool[1048576], *root;
    int nMem, n_pattern;
    Node* new_Node() {
        pool[nMem] = Node();
        return &pool[nMem++];
    }
    void init() {
        nMem = 0;
        root = new_Node();
        n_pattern = 0;
        add("");
    }
    void add(const string& str) { insert(root, str, 0); }
    void insert(Node* cur, const string& str, int pos) {
        for (int i = pos; i < str.size(); i++) {
            if (!cur->go[str[i] - 'a'])
                cur->go[str[i] - 'a'] = new_Node();
            cur = cur->go[str[i] - 'a'];
        }
        cur->cnt++;
        cur->i = n_pattern++;
    }
    void make_fail() {
        queue<Node*> que;
        que.push(root);
        while (!que.empty()) {
            Node* fr = que.front();
            que.pop();
            for (int i = 0; i < 26; i++) {
                if (fr->go[i]) {
                    Node* ptr = fr->fail;
                    while (ptr && !ptr->go[i]) ptr = ptr->fail;
                    fr->go[i]->fail = ptr =
                        (ptr ? ptr->go[i] : root);
                    fr->go[i]->dic = (ptr->cnt ? ptr : ptr->dic);
                    que.push(fr->go[i]);
                }
            }
        }
    }
    void query(string s) {
        Node* cur = root;
        for (int i = 0; i < (int)s.size(); i++) {
            while (cur && !cur->go[s[i] - 'a']) cur = cur->fail;
            cur = (cur ? cur->go[s[i] - 'a'] : root);
            if (cur->i >= 0) ans[cur->i]++;
            for (Node* tmp = cur->dic; tmp; tmp = tmp->dic)
                ans[tmp->i]++;
        } // ans[i] : number of occurrence of pattern i
    }
} AC;
```

6.4 Suffix Automata

```
/*最多有2n-1個狀態、3n-4個轉移，
parent tree上父節點是子節點的后綴，且endpos集合=所有子節點
的endpos，
節點代表的字串個數是Len(v)-Len(Link(v))，
常用對Len大到小去拓排(葉到根)代替dfs*/
const int MXN = 200005;
struct SAM {
    struct State {
        int len, link, sz;
        int next[26];
    } st[MXN<<1];
    int sz, last;
    int cnt[MXN<<1], idx[MXN<<1];
    SAM() {
        sz = 1;
        last = 0;
        cnt[0] = 1;
        idx[0] = 0;
    }
    void add(char c) {
        int p = last, np = sz++;
        st[np].len = st[p].len + 1;
        st[np].link = p;
        while (p && !st[p].next[c])
            st[p].next[c] = np, p = st[p].link;
        if (!p)
            st[np].link = 0;
        else
            p = st[p].next[c];
        if (st[p].len == st[np].len)
            cnt[np] = cnt[p] + 1;
        else
            cnt[np] = cnt[p] - st[p].len + 1;
        idx[np] = idx[p] + 1;
        last = np;
    }
    int getLen(int v) {
        return st[v].len;
    }
    int getCnt(int v) {
        return cnt[v];
    }
    int getIdx(int v) {
        return idx[v];
    }
}
```

```

    st[0].len = 0;
    st[0].link = -1;
    st[0].sz = 0;
    memset(st[0].next, -1, sizeof(st[0].next));
    sz = 1;
    last = 0;
}
void push(string& s){
    for(auto c : s) add(c-'a');
}
void add(int ch) {
    int cur = sz++;
    st[cur].len = st[last].len + 1;
    st[cur].sz = 1;
    memset(st[cur].next, -1, sizeof(st[cur].next));
    int p = last;
    while (p != -1 && st[p].next[ch] == -1) {
        st[p].next[ch] = cur;
        p = st[p].link;
    }
    if(p != -1 && st[p].next[ch] != cur){
        int q = st[p].next[ch];
        if (st[p].len + 1 == st[q].len) {
            st[cur].link = q;
        }
        else {
            int clone = sz++;
            st[clone].len = st[p].len + 1;
            st[clone].link = st[q].link;
            st[clone].sz = 0;
            memcpy(st[clone].next, st[q].next, sizeof(
                st[q].next));
            while (p != -1 && st[p].next[ch] == q) {
                st[p].next[ch] = clone;
                p = st[p].link;
            }
            st[q].link = st[cur].link = clone;
        }
    }
    last = cur;
}
void calc_endpos(){ // counting sort + bfs
    int max_len = 0;
    for(int i = 0; i < sz; i++){
        max_len = max(max_len, st[i].len);
        cnt[i] = 0;
    }
    for(int i = 0; i < sz; i++) cnt[st[i].len]++;
    for(int i = 1; i <= max_len; i++) cnt[i] += cnt[i-1];
    for(int i = 0; i < sz; i++) idx[--cnt[st[i].len]] = i;
    for(int i = sz-1; i > 0; i--){
        int u = idx[i], f = st[u].link;
        if(f != -1) st[f].sz += st[u].sz;
    }
}
void trans(string& t){
    int cur = 0, step = 0;
    for(int i = 0; i < t.size(); i++){
        int ch = t[i] - 'a';
        while(cur != -1 && st[cur].next[ch] == -1){
            cur = st[cur].link;
            if(cur != -1) step = st[cur].len;
        }
        if(cur == -1) cur = 0, step = 0;
        else cur = st[cur].next[ch], step++;
    }
}
} sam;

```

6.5 Z-algorithm

```

int z[MAXN];
void Z_value(const string& s) {
    // z[i] = lcp(s[1...], s[i...])
    int i, j, left, right, len = s.size();
    left = right = 0;
    z[0] = len;
    for (i = 1; i < len; i++) {
        j = max(min(z[i - left], right - i), 0);
        for (; i + j < len && s[i + j] == s[j]; j++);
        z[i] = j;
        if (i + z[i] > right) {

```

```

            right = i + z[i];
            left = i;
        }
    }
}

```

6.6 Knuth-Morris-Pratt Algorithm

```

vector<int> KMP(string s, string t) {
    s = t + '@' + s;
    int sz = s.size();
    vector<int> pi(sz);
    for (int i = 1; i < sz; i++) {
        int len = pi[i - 1];
        while (len != 0 && s[i] != s[len]) len = pi[len - 1];
        if (s[i] == s[len]) pi[i] = len + 1;
    }
    return pi;
}

```

6.7 Manacher's Algorithm

```

vector<int> d1(n), d2(n);
for (int i = 0, l = 0, r = -1; i < n; i++) {
    int k = (i > r) ? 1 : min(d1[l + r - i], r - i + 1);
    while (0 <= i - k && i + k < n && s[i - k] == s[i + k])
        k++;
    d1[i] = k--;
    if (i + k > r) l = i - k, r = i + k;
}
for (int i = 0, l = 0, r = -1; i < n; i++) {
    int k = (i > r) ? 0 : min(d2[l + r - i + 1], r - i + 1);
    while (0 <= i - k - 1 && i + k < n &&
        s[i - k - 1] == s[i + k])
        k++;
    d2[i] = k--;
    if (i + k > r) l = i - k - 1, r = i + k;
}

```

6.8 Minimum Rotation

```

int min_rotation(string s) {
    int a = 0, N = s.size();
    s += s;
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            if (a + j == i || s[a + j] < s[i + j]) {
                i += max(0, j - 1);
                break;
            }
            if (s[a + j] > s[i + j]) {
                a = i;
                break;
            }
        }
    }
    return a;
}
// rotate(begin(s), begin(s)+minRotation(s), end(s))
// 上面註解可以直接讓一個字串做 rotate，得到字典序最小的
// rotation

```

6.9 Rolling Hash

```

ll P1 = 75577, P2 = 12721, MOD = 998244353;
pii hashes[MAXN], hashes2[MAXN];
string s1, s2;
void build(string s, pii hash) {
    ll val1 = 0, val2 = 0;
    for (int i = 0; i < s.length(); i++) {
        val1 = (val1 * P1 % MOD + s[i]) % MOD;
        val2 = (val2 * P2 % MOD + s[i]) % MOD;
        hash[i] = {val1, val2};
    }
}
ll minus(ll a, ll b) { return (a - b + MOD) % MOD; }
ll cnt_occur() { // 檢查
    int n = s1.size(), k = s2.size();
    ll cnt = (hashes2[k - 1].x == hashes[k - 1].x &&
        hashes2[k - 1].y == hashes[k - 1].y);
    ll val1 = 0, val2 = 0;
    for (int i = 1, j = k; j < n; j++, i++) {
        val1 = minus(hashes[j].x,
            hashes[i - 1].x * fpow(P1, j - i + 1));

```

```

    val2 = minus(hashes[j].y,
        hashes[i - 1].y * fpow(P1, j - i + 1));
    if (hashs2[k - 1].x == val1 &&
        hashes2[k - 1].y == val2)
        cnt++;
}
return cnt;
}

```

6.10 Another Hash

```

const int M = 1e9 + 7;
const int C = 13331;
struct HashString { // 1-base
    int n;
    vector<ll> pows, sums;
    HashString(string s)
        : n(s.size()), pows(n + 1, 1), sums(n + 1) {
        for (int i = 1; i <= n; i++) {
            pows[i] = pows[i - 1] * C % M;
            sums[i] = (sums[i - 1] * C + s[i - 1]) % M;
        }
    }
    // Returns the hash of the substring (l, r]
    ll hash(int l, int r) {
        ll h = sums[r] - sums[l] * pows[r - l];
        return (h % M + M) % M;
    }
};

```

6.11 字典樹

```

struct Trie{
    struct node{
        node *nxt[26];
        int cnt, sz;
        node():cnt(0),sz(0){
            memset(nxt,0,sizeof(nxt));
        }
    };
    node *root;
    Trie(){ root = new node();}
    void insert(const string& s){
        node *now = root;
        for(auto i:s){
            now->sz++;
            if(now->nxt[i-'a'] == NULL){
                now->nxt[i-'a'] = new node();
            }
            now = now->nxt[i-'a'];
        }
        now->cnt++;
        now->sz++;
    }
} trie;

```

7 Geometry

7.1 Point

```

using ld = long double;
template<class T>
struct pt{
    T x,y;
    pt(T _x,T _y):x(_x),y(_y){}
    pt():x(0),y(0){}
    pt operator * (T c){ return pt(x*c,y*c);}
    pt operator / (T c){ return pt(x/c,y/c);}
    pt operator + (pt a){ return pt(x+a.x,y+a.y);}
    pt operator - (pt a){ return pt(x-a.x,y-a.y);}
    T operator * (pt a){ return x*a.x + y*a.y;}
    T operator ^ (pt a){ return x*a.y - y*a.x;}

    auto operator<=>(pt o) const { return (x != o.x) ? x <= o.x : y <= o.y; } // c++20
    bool operator < (pt a) const { return x < a.x || (x == a.x && y < a.y);};
    bool operator== (pt a) const { return x == a.x and y == a.y;};
    friend T ori(pt a, pt b, pt c) { return (b - a) ^ (c - a); }
    friend T abs2(pt a) { return a * a; }
};

```

```

// using numbers::pi; // c++20
const ld pi = acos(-1);
const ld eps = 1e-8L;
using Pt = pt<ld>;
int sgn(ld x) { return (x > -eps) - (x < eps); } // dcmp
    == sgn
ld abs(Pt a) { return sqrt(abs2(a)); }
ld arg(Pt x) { return atan2(x.y, x.x); }
bool argcmp(Pt a, Pt b) { // arg(a) < arg(b)
    int f = (Pt{a.y, -a.x} > Pt{}) ? 1 : -1 * (a != Pt{});
    int g = (Pt{b.y, -b.x} > Pt{}) ? 1 : -1 * (b != Pt{});
    return f == g ? (a ^ b) > 0 : f < g;
}
Pt unit(Pt x) { return x / abs(x); }
Pt rotate(Pt u) { // pi / 2
    return {-u.y, u.x};
}
Pt rotate(Pt u, ld a) {
    Pt v{sin(a), cos(a)};
    return {u ^ v, u * v};
}

istream &operator>>(istream &s, Pt &a) { return s >> a.x >> a.y; }
ostream &operator<<(ostream &s, Pt &a) { return s << "(" << a.x << ", " << a.y << ")";}

bool collinearity(Pt a, Pt b, Pt c) { // 三點共線
    return ((b - a) ^ (c - a)) == 0;
}

```

7.2 Line

```

struct Line {
    Pt a, b;
    Pt dir() { return b - a; }
};
int PtSide(Pt p, Line l) {
    return sgn(ori(l.a, l.b, p));
}
bool PtOnSeg(Pt p, Line l) {
    return PtSide(p, l) == 0 and sgn((p - l.a) * (p - l.b)) <= 0;
}
Pt proj(Pt p, Line l) {
    Pt dir = unit(l.b - l.a);
    return l.a + dir * (dir * (p - l.a));
}

```

7.3 Circle

```

struct Cir {
    Pt o;
    ld r;
};
bool disjunct(const Cir &a, const Cir &b) { // 分開 // 外切 -> 1
    return sgn(abs(a.o - b.o) - a.r - b.r) >= 0;
}
bool contain(const Cir &a, const Cir &b) {
    return sgn(a.r - b.r - abs(a.o - b.o)) >= 0;
}

```

7.4 點線距

```

ld PtSegDist(Pt p, Line l) {
    ld ans = min(abs(p - l.a), abs(p - l.b));
    if (sgn(abs(l.a - l.b)) == 0) return ans;
    if (sgn((l.a - l.b) * (p - l.b)) < 0) return ans;
    if (sgn((l.b - l.a) * (p - l.a)) < 0) return ans;
    return min(ans, abs(ori(p, l.a, l.b)) / abs(l.a - l.b));
}
ld SegDist(Line l, Line m) {
    return PtSegDist({0, 0}, {l.a - m.a, l.b - m.b});
}

```

7.5 PointInPolygon

```

int inPoly(Pt p, const vector<Pt> &P) {
    const int n = P.size();
    int cnt = 0;
    for (int i = 0; i < n; i++) {
        Pt a = P[i], b = P[(i + 1) % n];
    }
}

```

```

    if (PtOnSeg(p, {a, b})) return 1; // on edge
    if ((sgn(a.y - p.y) == 1) ^ (sgn(b.y - p.y) == 1))
        cnt += sgn(ori(a, b, p));
}
return cnt == 0 ? 0 : 2; // out, in
}

```

7.6 Intersection Of Line

```

bool isInter(Line l, Line m) {
    if (PtOnSeg(m.a, l) or PtOnSeg(m.b, l) or
        PtOnSeg(l.a, m) or PtOnSeg(l.b, m))
        return true;
    return PtSide(m.a, l) * PtSide(m.b, l) < 0 and
        PtSide(l.a, m) * PtSide(l.b, m) < 0;
}

Pt LineInter(Line l, Line m) {
    double s = ori(m.a, m.b, l.a), t = ori(m.a, m.b, l.b);
    return (l.b * s - l.a * t) / (s - t);
}

bool isInter(Line l, Line m) {
    if (PtOnSeg(m.a, l) or PtOnSeg(m.b, l) or
        PtOnSeg(l.a, m) or PtOnSeg(l.b, m))
        return true;
    return PtSide(m.a, l) * PtSide(m.b, l) < 0 and
        PtSide(l.a, m) * PtSide(l.b, m) < 0;
}

Pt LineInter(Line l, Line m) {
    double s = ori(m.a, m.b, l.a), t = ori(m.a, m.b, l.b);
    return (l.b * s - l.a * t) / (s - t);
}

```

7.7 Intersection Of Circle

```

vector<Pt> CircleInter(Cir a, Cir b) {
    double d2 = abs2(a.o - b.o), d = sqrt(d2);
    if (d < max(a.r, b.r) - min(a.r, b.r) || d > a.r + b.r)
        return {};
    Pt u = (a.o + b.o) / 2 + (a.o - b.o) * ((b.r * b.r - a.r * a.r) / (2 * d2));
    double A = sqrt((a.r + b.r + d) * (a.r - b.r + d) * (a.r + b.r - d) * (-a.r + b.r + d));
    Pt v = rotate(b.o - a.o) * A / (2 * d2);
    if (sgn(v.x) == 0 and sgn(v.y) == 0) return {u};
    return {u - v, u + v}; // counter clockwise of a
}

```

7.8 Intersection Of Circle and Line

```

vector<Pt> CircleLineInter(Cir c, Line l) {
    Pt H = proj(c.o, l);
    Pt dir = unit(l.b - l.a);
    double h = abs(H - c.o);
    if (sgn(h - c.r) > 0) return {};
    double d = sqrt(max((double)0., c.r * c.r - h * h));
    if (sgn(d) == 0) return {H};
    return {H - dir * d, H + dir * d};
    // Counterclockwise
}

```

7.9 Convex Hull

```

vector<Pt> Hull(vector<Pt> P) {
    sort(all(P));
    P.erase(unique(all(P)), P.end());
    P.insert(P.end(), P.rbegin() + 1, P.rend());
    vector<Pt> stk;
    for (auto p : P) {
        auto it = stk.rbegin();
        while (stk.rend() - it >= 2 and \
            ori(*next(it), *it, p) <= 0 and \
            (*next(it) < *it) == (*it < p)) {
            it++;
        }
        stk.resize(stk.rend() - it);
        stk.push_back(p);
    }
    stk.pop_back();
    return stk;
}

```

7.10 旋轉卡尺

```

auto RotatingCalipers(const vector<Pt> &hull) { // 最遠點
    對 回傳距離平方
}

```

```

int n = hull.size();
auto ret = abs2(hull[0]);
ret = 0;
if (hull.size() <= 2) return abs2(hull[0] - hull[1]);
for (int i = 0, j = 2; i < n; i++) {
    Pt a = hull[i], b = hull[(i + 1) % n];
    while(ori(hull[j], a, b) <
        (ori(hull[(j + 1) % n], a, b)))
        j = (j + 1) % n;
    chmax(ret, abs2(a - hull[j]));
    chmax(ret, abs2(b - hull[j]));
}
return ret;
}

```

7.11 Convex Hull Trick

```

struct Convex {
    int n;
    vector<Pt> A, V, L, U;
    Convex(const vector<Pt> &_A) : A(_A), n(_A.size()) {
        // n >= 3
        auto it = max_element(all(A));
        L.assign(A.begin(), it + 1);
        U.assign(it, A.end()), U.push_back(A[0]);
        for (int i = 0; i < n; i++) {
            V.push_back(A[(i + 1) % n] - A[i]);
        }
    }

    int inside(Pt p, const vector<Pt> &h, auto f) {
        auto it = lower_bound(all(h), p, f);
        if (it == h.end()) return 0;
        if (it == h.begin()) return p == *it;
        return 1 - sgn(ori(*prev(it), p, *it));
    }

    // 0: out, 1: on, 2: in
    int inside(Pt p) {
        return min(inside(p, L, less{}), inside(p, U, greater{}));
    }

    static bool cmp(Pt a, Pt b) { return sgn(a ^ b) > 0; }
    // A[i] is a far/closer tangent point
    int tangent(Pt v, bool close = true) {
        assert(v != Pt{});
        auto l = V.begin(), r = V.begin() + L.size() - 1;
        if (v < Pt{}) l = r, r = V.end();
        if (close) return (lower_bound(l, r, v, cmp) - V.begin()) % n;
        return (upper_bound(l, r, v, cmp) - V.begin()) % n;
    }

    // closer tangent point array[0] -> array[1] 順時針
    array<int, 2> tangent2(Pt p) {
        array<int, 2> t{-1, -1};
        if (inside(p) == 2) return t;
        if (auto it = lower_bound(all(L), p); it != L.end()
            and p == *it) {
            int s = it - L.begin();
            return {(s + 1) % n, (s - 1 + n) % n};
        }
        if (auto it = lower_bound(all(U), p, greater{});
            it != U.end() and p == *it) {
            int s = it - U.begin() + L.size() - 1;
            return {(s + 1) % n, (s - 1 + n) % n};
        }
        for (int i = 0; i != t[0]; i = tangent((A[t[0] = i] - p), 0));
        for (int i = 0; i != t[1]; i = tangent((p - A[t[1] = i]), 1));
        return t;
    }

    int find(int l, int r, Line L) {
        if (r < l) r += n;
        int s = PtSide(A[l % n], L);
        return *ranges::partition_point(views::iota(l, r), [&](int m) {
            return PtSide(A[m % n], L) == s;
        }) - l;
    };

    // Line A_x A_{x+1} intersect with L
    vector<int> intersect(Line L) {
        int l = tangent(L.a - L.b), r = tangent(L.b - L.a);
    }
}

```

```

    if (PtSide(A[l], L) * PtSide(A[r], L) >= 0) return
    {}
    return {find(l, r, L) % n, find(r, l, L) % n};
}
};

```

7.12 Area Of Circle and Polygon

```

double CirclePoly(Cir C, const vector<Pt> &P) {
    auto arg = [&](Pt p, Pt q) { return atan2(p ^ q, p * q
    ); };
    double r2 = C.r * C.r / 2;
    auto tri = [&](Pt p, Pt q) {
        Pt d = q - p;
        auto a = (d * p) / abs2(d), b = (abs2(p) - C.r * C
        .r) / abs2(d);
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a - sqrt(det)), t = min(1., -a +
        sqrt(det));
        if (t < 0 or 1 <= s) return arg(p, q) * r2;
        Pt u = p + d * s, v = p + d * t;
        return arg(p, u) * r2 + (u ^ v) / 2 + arg(v, q) *
        r2;
    };
    double sum = 0.0;
    for (int i = 0; i < P.size(); i++)
        sum += tri(P[i] - C.o, P[(i + 1) % P.size()] - C.o);
    return sum;
}

```

7.13 Area Of Circle and Triangle

```

double CircleTriangle(Pt a, Pt b, double r) {
    if (sgn(abs(a) - r) <= 0 and sgn(abs(b) - r) <= 0) {
        return abs(a ^ b) / 2;
    }
    if (abs(a) > abs(b)) swap(a, b);
    auto I = CircleLineInter({{a, b}}, r, {a, b});
    erase_if(I, [&](Pt x) { return !PtOnSeg(x, {a, b}); });
    if (I.size() == 1) return abs(a ^ I[0]) / 2 +
        SectorArea(I[0], b, r);
    if (I.size() == 2) {
        return SectorArea(a, I[0], r) + SectorArea(I[1], b
        , r) + abs(I[0] ^ I[1]) / 2;
    }
    return SectorArea(a, b, r);
}

```

7.14 Area Of Sector

$$\angle AOB * r^2 / 2$$

```

double Sector(Pt a, Pt b, double r) {
    double theta = atan2(a.y, a.x) - atan2(b.y, b.x);
    while (theta <= 0) theta += 2 * pi;
    while (theta >= 2 * pi) theta -= 2 * pi;
    theta = min(theta, 2 * pi - theta);
    return r * r * theta / 2;
}

```

7.15 切線

```

vector<Line> CircleTangent(Cir c, Pt p) {
    vector<Line> z;
    double d = abs(p - c.o);
    if (sgn(d - c.r) == 0) {
        Pt i = rotate(p - c.o);
        z.push_back({p, p + i});
    } else if (d > c.r) {
        double o = acos(c.r / d);
        Pt i = unit(p - c.o);
        Pt j = rotate(i, o) * c.r;
        Pt k = rotate(i, -o) * c.r;
        z.push_back({c.o + j, p});
        z.push_back({c.o + k, p});
    }
    return z;
}

```

7.16 公切線

```

vector<Line> CircleTangent(Cir c1, Cir c2, int sign1) {
    // sign1 = 1 for outer tang, -1 for inner tang

```

```

    vector<Line> ret;
    ld d_sq = abs2(c1.o - c2.o);
    if (sgn(d_sq) == 0) return ret;
    ld d = sqrt(d_sq);
    Pt v = (c2.o - c1.o) / d;
    ld c = (c1.r - sign1 * c2.r) / d;
    if (c * c > 1) return ret;
    ld h = sqrt(max(0.0, 1.0 - c * c));
    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
        Pt n = Pt(v.x * c - sign2 * h * v.y, v.y * c +
        sign2 * h * v.x);
        Pt p1 = c1.o + n * c1.r;
        Pt p2 = c2.o + n * (c2.r * sign1);
        if (sgn(p1.x - p2.x) == 0 && sgn(p1.y - p2.y) ==
        0)
            p2 = p1 + rotate(c2.o - c1.o);
        ret.push_back({p1, p2});
    }
    return ret;
}

```

7.17 半平面交

```

bool cover(Line L, Line P, Line Q) {
    // PtSide(LineInter(P, Q), L) <= 0 or P, Q parallel
    i128 u = (Q.a - P.a) ^ Q.dir();
    i128 v = P.dir() ^ Q.dir();
    i128 x = P.dir().x * u + (P.a - L.a).x * v;
    i128 y = P.dir().y * u + (P.a - L.a).y * v;
    return sgn(x * L.dir().y - y * L.dir().x) * sgn(v) >=
    0;
}
vector<Line> HPI(vector<Line> P) {
    // Line P.a -> P.b 的逆時針是半平面
    sort(all(P), [&](Line l, Line m) {
        if (argcmp(l.dir(), m.dir())) return true;
        if (argcmp(m.dir(), l.dir())) return false;
        return ori(m.a, m.b, l.a) > 0;
    });
    int n = P.size(), l = 0, r = -1;
    for (int i = 0; i < n; i++) {
        if (i and !argcmp(P[i - 1].dir(), P[i].dir()))
            continue;
        while (l < r and cover(P[i], P[r - 1], P[r])) r--;
        while (l < r and cover(P[i], P[l], P[l + 1])) l++;
        P[++r] = P[i];
    }
    while (l < r and cover(P[l], P[r - 1], P[r])) r--;
    while (l < r and cover(P[r], P[l], P[l + 1])) l++;
    if (r - l <= 1 or !argcmp(P[l].dir(), P[r].dir()))
        return {}; // empty
    if (cover(P[l + 1], P[l], P[r]))
        return {}; // infinity
    return vector(P.begin() + l, P.begin() + r + 1);
}

```

7.18 最小包圍圓

```

Pt Center(Pt a, Pt b, Pt c) {
    Pt x = (a + b) / 2;
    Pt y = (b + c) / 2;
    return LineInter({x, x + rotate(b - a)}, {y, y +
    rotate(c - b)});
}
Cir MEC(vector<Pt> P) {
    mt19937 rng(time(0));
    shuffle(all(P), rng);
    Cir C = {P[0], 0.0};
    for (int i = 0; i < P.size(); i++) {
        if (C.inside(P[i])) continue;
        C = {P[i], 0};
        for (int j = 0; j < i; j++) {
            if (C.inside(P[j])) continue;
            C = {(P[i] + P[j]) / 2, abs(P[i] - P[j]) / 2};
            for (int k = 0; k < j; k++) {
                if (C.inside(P[k])) continue;
                C.o = Center(P[i], P[j], P[k]);
                C.r = abs(C.o - P[i]);
            }
        }
    }
    return C;
}

```

7.19 圓聯集面積

```
// Area[i] : area covered by at least i circle
// TODO:!!!!aa!!!
vector<double> CircleUnion(const vector<Cir> &C) {
    const int n = C.size();
    vector<double> Area(n + 1);
    auto check = [&](int i, int j) {
        if (!contain(C[i], C[j]))
            return false;
        return sgn(C[i].r - C[j].r) > 0 or (sgn(C[i].r - C[j].r) == 0 and i < j);
    };
    struct Teve {
        double ang; int add; Pt p;
        bool operator<(const Teve &b) { return ang < b.ang; }
    };
    auto ang = [&](Pt p) { return atan2(p.y, p.x); };
    for (int i = 0; i < n; i++) {
        int cov = 1;
        vector<Teve> event;
        for (int j = 0; j < n; j++) if (i != j) {
            if (check(j, i)) cov++;
            else if (!check(i, j) and !disjunct(C[i], C[j])) {
                auto I = CircleInter(C[i], C[j]);
                assert(I.size() == 2);
                double a1 = ang(I[0] - C[i].o), a2 = ang(I[1] - C[i].o);
                event.push_back({a1, 1, I[0]});
                event.push_back({a2, -1, I[1]});
                if (a1 > a2) cov++;
            }
        }
        if (event.empty()) {
            Area[cov] += pi * C[i].r * C[i].r;
            continue;
        }
        sort(all(event));
        event.push_back(event[0]);
        for (int j = 0; j + 1 < event.size(); j++) {
            cov += event[j].add;
            Area[cov] += (event[j].p ^ event[j + 1].p) / 2.;
            double theta = event[j + 1].ang - event[j].ang;
            if (theta < 0) theta += 2 * pi;
            Area[cov] += (theta - sin(theta)) * C[i].r * C[i].r / 2.;
        }
    }
    return Area;
}
```

7.20 多邊形聯集面積

```
// Area[i] : area covered by at least i polygon
vector<double> PolyUnion(const vector<vector<Pt>> &P) {
    const int n = P.size();
    vector<double> Area(n + 1);
    vector<Line> Ls;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < P[i].size(); j++)
            Ls.push_back({P[i][j], P[i][(j + 1) % P[i].size()]});
    auto cmp = [&](Line &l, Line &r) {
        Pt u = l.b - l.a, v = r.b - r.a;
        if (argcmp(u, v)) return true;
        if (argcmp(v, u)) return false;
        return PtSide(l.a, r) < 0;
    };
    sort(all(Ls), cmp);
    for (int l = 0, r = 0; l < Ls.size(); l = r) {
        while (r < Ls.size() and !cmp(Ls[l], Ls[r])) r++;
        Line L = Ls[l];
        vector<pair<Pt, int>> event;
        for (auto [c, d] : Ls) {
            if (sgn((L.a - L.b) ^ (c - d)) != 0) {
                int s1 = PtSide(c, L) == 1;
                int s2 = PtSide(d, L) == 1;
                if (s1 ^ s2) event.emplace_back(LineInter(L, {c, d}), s1 ? 1 : -1);
            }
        }
    }
}
```

```
} else if (PtSide(c, L) == 0 and sgn((L.a - L.b) * (c - d)) > 0) {
    event.emplace_back(c, 2);
    event.emplace_back(d, -2);
}
}
sort(all(event), [&](auto i, auto j) {
    return (L.a - i.ff) * (L.a - L.b) < (L.a - j.ff) * (L.a - L.b);
});
int cov = 0, tag = 0;
Pt lst{0, 0};
for (auto [p, s] : event) {
    if (cov >= tag) {
        Area[cov] += lst ^ p;
        Area[cov - tag] -= lst ^ p;
    }
    if (abs(s) == 1) cov += s;
    else tag += s / 2;
    lst = p;
}
for (int i = n - 1; i >= 0; i--) Area[i] += Area[i + 1];
for (int i = 1; i <= n; i++) Area[i] /= 2;
return Area;
};
```

7.21 Pick's Theory

$$A = i + \frac{b}{2} - 1, \quad 2A = 2i + b - 2, \quad i = A - \frac{b}{2} + 1$$

其中 A 為面積、 b 為圖形內部的點數量、 i 為圖形線上的點數量

```
vector<Pt> p;
ll compute_area(int n) { // A * 2
    ll area = 0;
    for (int i = 1; i < n; i++)
        area += cross(p[i], p[(i + 1) % n], p[0]);
    area = abs(area);
    return area;
}
ll compute_pt_on_line(int n) { // b
    ll pt = 0;
    for (int i = 0; i < n; i++)
        pt += __gcd(abs(p[i].x - p[(i + 1) % n].x), abs(p[i].y - p[(i + 1) % n].y));
    return pt; // 這是網路上找的
}
ll compute_inner_pt(int n) { // i
    return (compute_area(n) - compute_pt_on_line(n)) / 2 + 1;
}
```

8 DP

8.1 區間 DP

時間複雜度 $O(n^3)$ ，常見的 n 在 500 以內，至多 1,000。

```
// 區間大小為 1 為 base case
for (int i = 1; i <= n; i++) dp[i][i] = 0;
// 由小區間往大開始轉移
for (int len = 2; len <= n; len++)
    for (int l = 1, r = len; r <= n; l++, r++) {
        dp[l][r] = INF;
        for (int k = l; k < r; k++)
            dp[l][r] = min(dp[l][r], dp[l][k] + dp[k + 1][r] + n[i] * m[k] * n[r]);
    }
return dp[1][n];
```

8.2 位數 DP

dp[位數][狀態]，dp[pos][state]：
定義為目前位數在前導狀態為 state 的時候的計數
例如：求數字沒有出現 66 的數量 1 ~ r
→ dp[pos][1] 可表示計算 pos 個位數在前導出現一個 6 的計數
→ dp[3][1] 則計算 6xxx
模板的 pos 是反過來的，但不影響 (只是用來 dp 記憶用)
pos: 目前位數
state: 前導狀態
lead: 是否有前導 0
(大部分題目不用但有些數字 e.g. 00146 如果有影響時要考慮)
limit: 是否窮舉有被 num 限制

```
vector<int> num;
int dp[20][state];
int dfs(int pos, int state, bool lead, bool limit) {
    if (pos == num.size()) { // 有時要根據不同state回傳情況
        return 1;
    }
    if (limit == false && lead == false &&
        dp[pos][state] != -1)
        return dp[pos][state];
    int up = limit ? num[pos] : 9;
    int ans = 0;
    for (int i = 0; i <= up; i++) {
        // 有時要考慮那些狀況要continue
        ans += dfs(pos + 1, state || (check[i] == 2),
                    lead && i == 0, limit && i == num[pos]);
    }
    if (limit == false && lead == false)
        dp[pos][state] = ans;
    return ans;
}
```

8.3 SOS DP

```
for (int mask = 0; mask < (1 << N); mask++)
    F[mask] = A[mask];
for (int i = 0; i < N; i++)
    for (int mask = 0; mask < (1 << N); mask++)
        if (mask & (1 << i)) F[mask] += F[mask ^ (1 << i)];
```

9 Sorting

9.1 Merge Sort

```
template <typename T>
void merge_sort(T &arr, T &tmp, int l, int r) {
    // 陣列元素少於 1 直接結束
    if (r <= l + 1) return;

    // 分成左右兩邊處理
    int m = l + (r - l) / 2;
    merge_sort(arr, tmp, l, m); // [l, m)
    merge_sort(arr, tmp, m, r); // [m, r)

    // 合併兩邊的結果
    for (int i = l, li = l, ri = m; i < r; i++) {
        // 只剩下右邊 // 兩邊都有剩，右邊比左邊小
        if (li == m || (ri != r && arr[ri] < arr[li]))
            tmp[i] = arr[ri++];
        // 只剩下左邊 // 兩邊都有剩，左邊比右邊小
        else tmp[i] = arr[li++];
    }

    // 把結果複製回來
    for (int i = l; i < r; i++) arr[i] = tmp[i];
}
```

9.2 Quick Sort

```
template <typename T>
void quick_sort(T arr[], int l, int r) {
    // 陣列元素少於 1 直接結束
    if (r <= l + 1)
        return;

    // 左邊和右邊的指針
    int li = l + 1, ri = r - 1;
    // 判定的基準點
    T pivot = arr[l];

    // TODO: 左邊 <= 基準點 < 右邊
    while (li != ri) {
        // 右邊的指針往左移動，找比基準點小的值
        while (arr[ri] > pivot && li < ri)
            ri--;
        // 左邊的指針往右移動，找比基準點大的值
        while (arr[li] <= pivot && li < ri)
            li++;
        // 當左右指針沒有相遇時，表示不滿足 TODO
        if (li < ri)
            swap(arr[li], arr[ri]);
    }
}
```

```
// 將基準點移到指針相遇的地方
arr[l] = arr[li];
arr[li] = pivot;

quick_sort(arr, l, li); // 處理基準點的左邊
quick_sort(arr, li + 1, r); // 處理基準點的右邊
}
```

10 Miscellaneous

10.1 Discretization

```
int arr[MXN], tmp[MXN], n;
// arr[i] 是初始陣列，長度為n，且 0-base
void discretization() {
    for (int i = 0; i < n; i++) // 將 arr 複製到 tmp
        tmp[i] = arr[i];
    sort(tmp, tmp + n); // 排序
    int len = unique(tmp, tmp + n) - (tmp); // 把 tmp 裡
    // 重複的數字去掉
    for (int i = 0; i < n; i++)
        arr[i] = lower_bound(tmp, tmp + len, arr[i]) - tmp;
    // 二分搜
}
```

10.2 Mo's Algorithm

```
struct Query {
    int l, r, id; // 左界、右界、編號
    Query(int _l, int _r, int _id)
        : l(_l), r(_r), id(_id) {}
    friend bool operator<(const Query &lhs,
                          const Query &rhs) {
        if (lhs.l / k == rhs.l / k) // 如果在同一塊就比較 r
            return lhs.r < rhs.r;
        return lhs.l < rhs.l; // 如果在不同快就比較 l
    }
};

vector<Query> q; // 詢問
void add(int idx) {} // ... 自己推喔
void sub(int idx) {} // ... 自己推喔

void mos_algorithm() {
    k = sqrt(n); // n 個東西分成 k 塊
    sort(q.begin(), q.end());
    for (int i = 0, l = 1, r = 0; i < m; i++) {
        while (l > q[i].l) add(--l);
        while (r < q[i].r) add(++r);
        while (l < q[i].l) sub(l--);
        while (r > q[i].r) sub(r--);
        ans[q[i].id] = cur; // 每一題長不一樣
    }
}
```

10.3 Decimal

```
from decimal import Decimal, getcontext, ROUND_FLOOR
getcontext().prec = 250 # set precision (MAX_PREC)
getcontext().Emax = 250 # set exponent limit (MAX_EMAX)
getcontext().rounding = ROUND_FLOOR # set round floor
itwo,two,N = Decimal(0.5),Decimal(2),200
pi = angle(Decimal(-1))
```

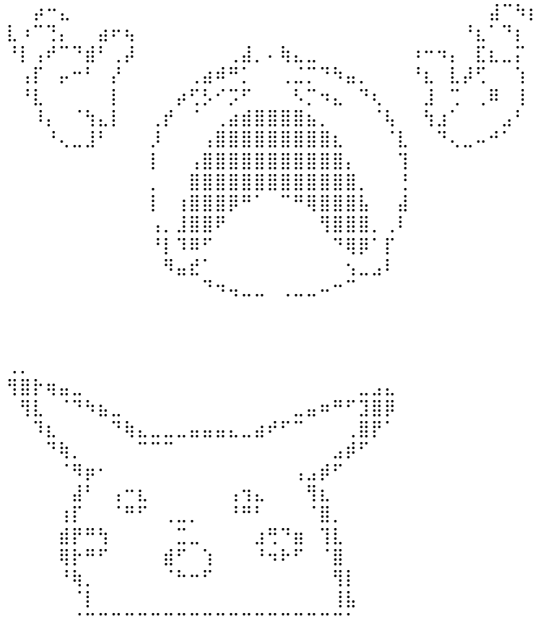
10.4 Fraction

```
from fractions import Fraction
import math
"""專門用來表示和操作有理數，可以進行算"""
frac1 = Fraction(1) # 1/1
frac2 = Fraction(1, 3) # 1/3
frac3 = Fraction(0.5) # 1/2
frac4 = Fraction('22/7') # 22/7
frac5 = Fraction(8, 16) # 自動約分為 1/2
frac9 = Fraction(22, 7)
frac9.numerator # 22
frac9.denominator # 7
x = Fraction(math.pi)
y2 = x.limit_denominator(100) # 分母限制為 100
print(y2) # 311/99
float(x) # 轉換為浮點數
```

```

//          !#####
//
//          !#####!          ##!
//
//          !#####!          ##
//
//          !#####          #####
//
//          #####          #####
//
//          !###!          !###!          #####
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//          ,#####'          #####! '          #####
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//          ,#####'          #####          !#####!
//
//          #####'          #####          #####
//
//          ~##
//
同志保佑 永無BUG

```



10.6 Graph Paper

